



OpenStack Labs

Lab 05: Deploying an External Instance

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About This Document

- This document was developed by a team at the University of Tennessee at Chattanooga led by Dr. Mengjun Xie (mengjun-xie@utc.edu).
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Introduction

In the previous lab, you created an external network, a router, and floating IP addresses. However, you were still not able to connect to an instance connected to the external network from outside the OpenStack environment. In this lab, you will create security groups to allow the traffic to reach the external instance, and you will create an SSH key pair so that you can remotely log in to the instance.

Objectives

- Create and manage security groups.
- Create and manage SSH key pairs.
- Launch and verify an external instance.

Lab Settings

The information in the table below will be needed in order to complete the lab. The task sections below provide details on the use of this information.

Virtual Machine	IP Address	Account	Password
workstation	ens3: 192.168.1.21 ens4: 172.25.250.21	ubuntu	ubuntu
devstack	ens3: 192.168.1.20 ens4: 172.25.250.20	ubuntu	ubuntu

1 Defining Security Groups

In this task, you will use the *Horizon Dashboard* and *OpenStack Unified CLI* to manage security groups for OpenStack instances. Security groups function like virtual firewalls, allowing you to define rules that allow or deny specific types of network traffic. Modifying the default security group settings is essential to enable communication with external instances from outside the OpenStack environment. In this lab, we are concerned with two types of traffic in particular: ICMP to allow the use of the **ping** command, and SSH to enable remote login from an external network.

- 1.1. Log in to the **workstation** machine as the **ubuntu** user with password **ubuntu**.

```
Ubuntu 18.04.6 LTS workstation tty1
workstation login: ubuntu
Password:
```

- 1.2. Launch the graphical user interface.

```
ubuntu@workstation:~$ startx
```

```
Welcome to Ubuntu 18.04.6 LTS (GNU/Linux 4.15.0-213-generic x86_64)

* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:        https://ubuntu.com/advantage

System information as of Fri Jun  7 21:01:55 UTC 2024

System load:  0.6               Processes:            197
Usage of /:   7.9% of 116.12GB  Users logged in:     0
Memory usage: 13%              IP address for ens3: 192.168.1.21
Swap usage:   0%               IP address for ens4: 172.25.250.21

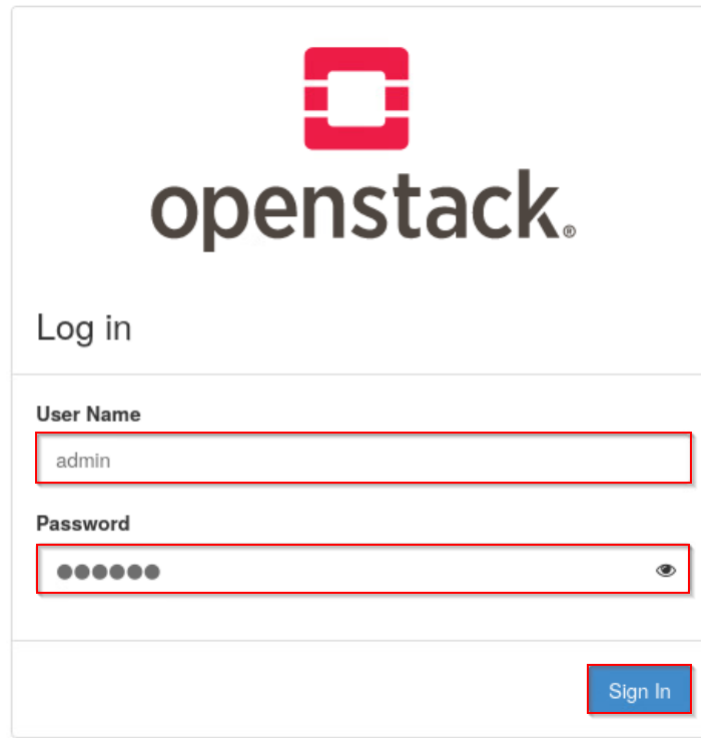
Expanded Security Maintenance for Infrastructure is not enabled.

2 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

146 additional security updates can be applied with ESM Infra.
Learn more about enabling ESM Infra service for Ubuntu 18.04 at
https://ubuntu.com/18-04

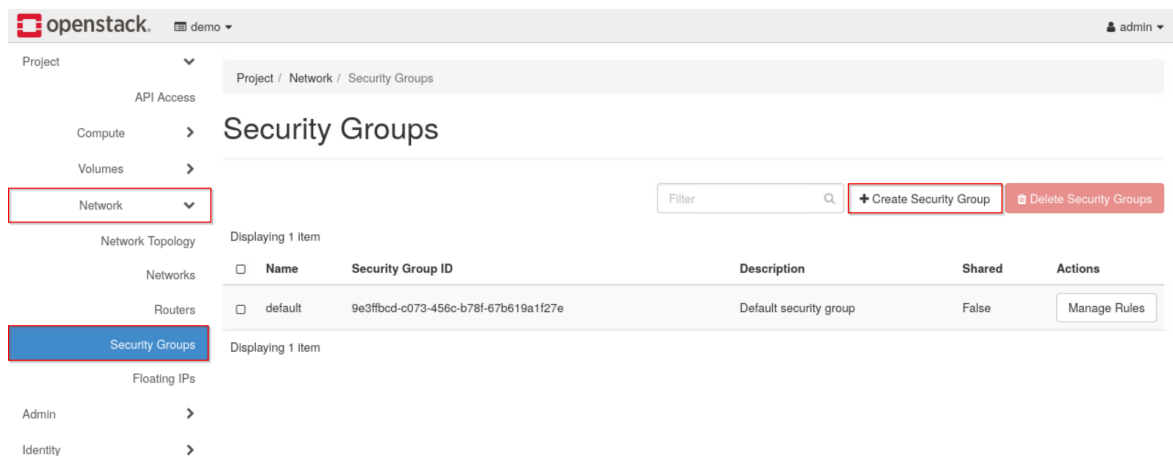
ubuntu@workstation:~$ startx_
```

- 1.3. Open the web browser. Navigate to **192.168.1.20**, and log in to the dashboard as **admin** with the password **secret**.



The image shows the OpenStack login page. At the top is the OpenStack logo, a red square with a white 'O' inside. Below the logo is the text 'openstack.' in a large, bold, sans-serif font. Underneath is the text 'Log in'. There are two input fields: 'User Name' with the text 'admin' and 'Password' with a masked password of seven dots. A blue 'Sign In' button is at the bottom right.

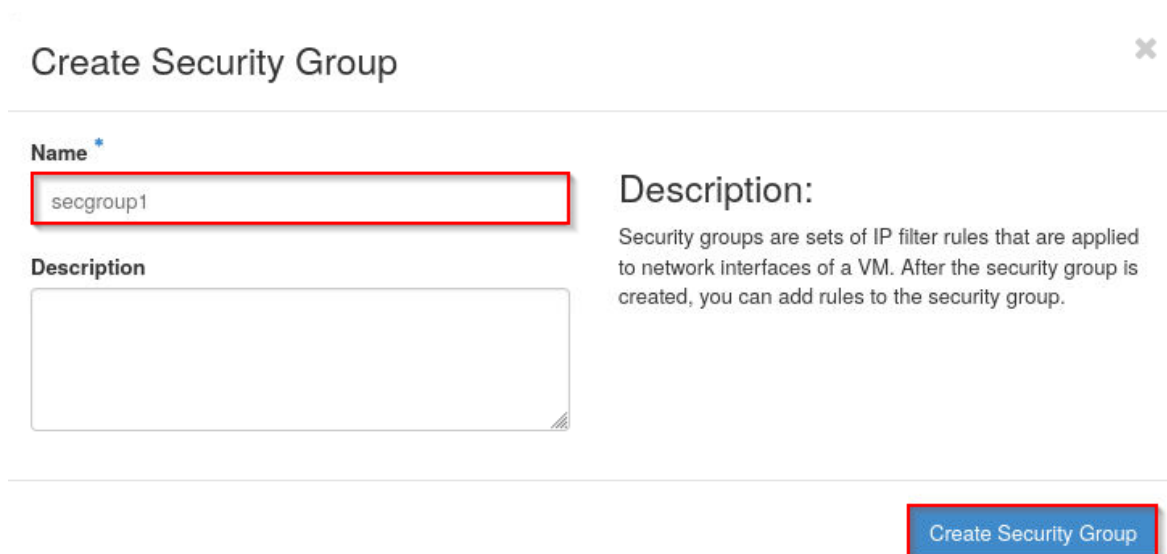
- 1.4. Select the **demo** project. Navigate to **Network > Security Groups** and click **Create Security Group**.



The image is a screenshot of the OpenStack dashboard. The top bar shows the OpenStack logo, 'demo' project, and 'admin' user. The left sidebar has a menu with 'Project', 'API Access', 'Compute', 'Volumes', 'Network', 'Network Topology', 'Networks', 'Routers', 'Security Groups', 'Floating IPs', 'Admin', and 'Identity'. 'Network' and 'Security Groups' are highlighted with red boxes. The main content area shows the 'Security Groups' page. It has a breadcrumb 'Project / Network / Security Groups' and a title 'Security Groups'. There are buttons for '+ Create Security Group' and 'Delete Security Groups'. Below is a table with one item:

Name	Security Group ID	Description	Shared	Actions
default	9e3ffbcd-c073-456c-b78f-67b619a1f27e	Default security group	False	Manage Rules

1.5. Enter **secgroup1** into the *Name* field and click **Create Security Group**.



Create Security Group

Name *

Description

Description:
 Security groups are sets of IP filter rules that are applied to network interfaces of a VM. After the security group is created, you can add rules to the security group.

Create Security Group

1.6. After creating the security group, you should be redirected to its rules page. If not, click **Manage Rules** in the same row as **secgroup1** on the **Security Groups** page to get there. From here, you can see that by default, the security group allows all egress (outgoing) traffic. No ingress rules are defined, so all incoming traffic is denied by default. Click **Add Rule** to add a new rule in the security group.

Project / Network / Security Groups / Manage Security Group Rules

Manage Security Group Rules: secgroup1
 (918d7354-9ec7-4102-8cf7-e87b5b4537c3)

+ Add Rule **Delete Rules**

Displaying 2 items

☐	Direction	Ether Type	IP Protocol	Port Range	Remote IP Prefix	Remote Security Group	Description	Actions
☐	Egress	IPv4	Any	Any	0.0.0.0/0	-	-	Delete Rule
☐	Egress	IPv6	Any	Any	:::0	-	-	Delete Rule

Displaying 2 items

- 1.7. Select **All ICMP** from the *Rule* dropdown and click **Add**. This allows ICMP traffic, including the **ping** command, to reach instances in this security group.

Add Rule ✕

Rule ^{*}

All ICMP ▼

Description [?]

Direction

Ingress ▼

Remote ^{*} [?]

CIDR ▼

CIDR ^{*} [?]

0.0.0.0/0

Description:

Rules define which traffic is allowed to instances assigned to the security group. A security group rule consists of three main parts:

Rule: You can specify the desired rule template or use custom rules, the options are Custom TCP Rule, Custom UDP Rule, or Custom ICMP Rule.

Open Port/Port Range: For TCP and UDP rules you may choose to open either a single port or a range of ports. Selecting the "Port Range" option will provide you with space to provide both the starting and ending ports for the range. For ICMP rules you instead specify an ICMP type and code in the spaces provided.

Remote: You must specify the source of the traffic to be allowed via this rule. You may do so either in the form of an IP address block (CIDR) or via a source group (Security Group). Selecting a security group as the source will allow any other instance in that security group access to any other instance via this rule.

Cancel

Add

Note

By default, *Direction* is set to **Ingress** and *CIDR* is set to **0.0.0.0/0**. **Ingress** specifies incoming traffic, and **0.0.0.0/0** specifies that the traffic is accepted from any IP address.

1.8. Click **Add Rule** again.

Project / Network / Security Groups / Manage Security Group Rules

Manage Security Group Rules: secgroup1 (918d7354-9ec7-4102-8cf7-e87b5b4537c3)

+ Add Rule
Delete Rules

Displaying 3 items

☐	Direction	Ether Type	IP Protocol	Port Range	Remote IP Prefix	Remote Security Group	Description	Actions
☐	Egress	IPv4	Any	Any	0.0.0.0/0	-	-	Delete Rule
☐	Egress	IPv6	Any	Any	:::0	-	-	Delete Rule
☐	Ingress	IPv4	ICMP	Any	0.0.0.0/0	-	-	Delete Rule

Displaying 3 items

- 1.9. Now, we will create a rule to allow SSH ingress traffic. Since SSH works over TCP, leave *Rule* as **Custom TCP Rule**. Under *Port*, enter **22**, which is the default SSH port. Click **Add** to add the rule.

Add Rule ✕

Rule *

Custom TCP Rule

Description ?

Direction

Ingress

Open Port *

Port

Port ?

22

Remote ?

CIDR

CIDR ?

0.0.0.0/0

Description:

Rules define which traffic is allowed to Instances assigned to the security group. A security group rule consists of three main parts:

Rule: You can specify the desired rule template or use custom rules, the options are Custom TCP Rule, Custom UDP Rule, or Custom ICMP Rule.

Open Port/Port Range: For TCP and UDP rules you may choose to open either a single port or a range of ports. Selecting the "Port Range" option will provide you with space to provide both the starting and ending ports for the range. For ICMP rules you instead specify an ICMP type and code in the spaces provided.

Remote: You must specify the source of the traffic to be allowed via this rule. You may do so either in the form of an IP address block (CIDR) or via a source group (Security Group). Selecting a security group as the source will allow any other Instance in that security group access to any other Instance via this rule.

Cancel
Add

- 1.10. Log out of the *Horizon Dashboard* and close the web browser.
- 1.11. If a terminal window is not already open, open one and source the **admin** credentials from the `~/keystonerc-admin` file.

```
ubuntu@workstation:~$ source ~/keystonerc-admin
```

```
ubuntu@workstation:~$ source ~/keystonerc-admin
[ubuntu@workstation (keystone-admin)]:~$
```

- 1.12. Before creating or modifying any security groups or rules, list the existing security groups to see what is already configured.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack security group list
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack security group list
```

ID	Name	Description	Project
2f0f5133-8396-45ea-a4de-61945d79ed2e	default	Default security group	eb2dcd08d8ae46ffac3f16c3973ef61d
62dafb67-e7fd-44d6-bf87-f4701dd66875	default	Default security group	39e851b14f864573aad60582c35e40dc
918d7354-9ec7-4102-8cf7-e87b5b4537c3	secgroup1		39e851b14f864573aad60582c35e40dc

```
[ubuntu@workstation (keystone-admin)]:~$
```

- 1.13. Now, we will recreate the **secgroup1** security group through the command line and add some additional rules necessary for connecting to an external instance. First, to demonstrate how to remove a rule from a security group, list the rules in the **secgroup1** security group and copy the ID of the ICMP rule.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack security group rule list \
> secgroup1
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack security group rule list \
> secgroup1
```

ID	IP Protocol	IP Range	Port Range	Remote Security Group
022a7e4f-41ce-4c37-9764-e22733a5bc19	None	None		None
8b0035c8-f81e-451d-bf0a-dd258ebca9f4	icmp	0.0.0.0/0		None
9ba01e49-f422-43e2-b731-8e25f58bc428	tcp	0.0.0.0/0	22:22	None
bce3af7e-2cee-48a3-99d2-3d05c709a159	None	None		None

```
[ubuntu@workstation (keystone-admin)]:~$
```

Tip

Since we want to copy the ID value, we do not want the output to fit the width of the terminal window, which would wrap the ID across multiple lines and prevent copying. To prevent this, maximize the terminal window before running this command.

Tip

To copy a value from the terminal, select the desired string with the mouse, then either right-click and click **Copy** or press **Ctrl+Shift+C**.

- 1.14.** Use the ID for the ICMP rule to delete that rule. Note that you do not have to list **secgroup1** in this command because IDs in OpenStack are *globally unique* within a deployment.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack security group rule delete \
> 8b0035c8-f81e-451d-bf08-dd258ebca9f4
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack security group rule delete \
> 8b0035c8-f81e-451d-bf0a-dd258ebca9f4
[ubuntu@workstation (keystone-admin)]:~$ █
```

Note

The actual ID value may differ.

Tip

To paste a value to the command line, either right-click and click **Paste** or press **Ctrl+Shift+V**.

- 1.15.** List the rules in the **secgroup1** security group again to ensure the rule was deleted successfully.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack security group rule list \
> secgroup1
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack security group rule list \
> secgroup1
```

ID	IP Protocol	IP Range	Port Range	Remote Security Group
022a7e4f-41ce-4c37-9764-e22733a5bc19	None	None		None
9ba01e49-f422-43e2-b731-8e25f58bc428	tcp	0.0.0.0/0	22:22	None
bce3af7e-2cee-48a3-99d2-3d05c709a159	None	None		None

```
[ubuntu@workstation (keystone-admin)]:~$ █
```

- 1.16.** Delete the **secgroup1** security group.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack security group delete \
> secgroup1
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack security group delete \
> secgroup1
[ubuntu@workstation (keystone-admin)]:~$ █
```

1.17. Create the **secgroup2** security group.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack security group create \
> secgroup2
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack security group create \
> secgroup2
+-----+-----+
| Field | Value |
+-----+-----+
| created_at | 2025-06-28T15:52:22Z |
| description | secgroup2 |
| id | d4792861-920e-4dd6-bdd1-e9ea3b5f6d6a |
| name | secgroup2 |
| project_id | 39e851b14f864573aad60582c35e40dc |
| revision_number | 1 |
| rules | created_at='2025-06-28T15:52:22Z', direction='egress', |
| | ethertype='IPv6', |
| | id='4ca422bf-c170-4159-b1e6-329c949d01e4', |
| | standard_attr_id='79', updated_at='2025-06-28T15:52:22Z' |
| | created_at='2025-06-28T15:52:22Z', direction='egress', |
| | ethertype='IPv4', |
| | id='c409c480-d296-4009-abf9-3e01c3b2c6ca', |
| | standard_attr_id='80', updated_at='2025-06-28T15:52:22Z' |
| updated_at | 2025-06-28T15:52:22Z |
+-----+-----+
[ubuntu@workstation (keystone-admin)]:~$
```

1.18. List the rules in the **secgroup2** security group. These are the default rules that exist upon creation.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack security group rule list \
> secgroup2
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack security group rule list \
> secgroup2
+-----+-----+-----+-----+-----+
| ID | IP Protocol | IP Range | Port Range | Remote Security Group |
+-----+-----+-----+-----+-----+
| 4ca422bf-c170-4159-b1e6-329c949d01e4 | None | None | | None |
| c409c480-d296-4009-abf9-3e01c3b2c6ca | None | None | | None |
+-----+-----+-----+-----+-----+
[ubuntu@workstation (keystone-admin)]:~$
```

Tip

You can use the command

```
openstack security group rule show <rule_id>
```

to show the details of each rule and confirm that they are the same default rules you get when creating a security group through the Horizon Dashboard. The rules allow all outgoing traffic over IPv4 and IPv6.

1.19. Add a security rule in the **secgroup2** security group to allow all incoming ICMP traffic.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack security group rule create \
> --protocol icmp \
> secgroup2
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack security group rule create \
> --protocol icmp \
> secgroup2
```

Field	Value
created_at	2025-06-28T16:05:07Z
description	
direction	ingress
ether_type	IPv4
id	822e6305-b7a4-4df6-8f7b-74d117171381
name	None
port_range_max	None
port_range_min	None
project_id	39e851b14f864573aad60582c35e40dc
protocol	icmp
remote_group_id	None
remote_ip_prefix	0.0.0.0/0
revision_number	0
security_group_id	d4792861-920e-4dd6-bdd1-e9ea3b5f6d6a
updated_at	2025-06-28T16:05:07Z

```
[ubuntu@workstation (keystone-admin)]:~$
```

Note

If no additional arguments are given, the direction defaults to **ingress** and the remote IP defaults to **0.0.0.0/0**. In other words, it allows all incoming traffic over the given protocol.

1.20. List the rules in the **secgroup2** security group again to ensure the ICMP rule was created successfully.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack security group rule list \
> secgroup2
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack security group rule list \
> secgroup2
```

ID	IP Protocol	IP Range	Port Range	Remote Security Group
4ca422bf-c170-4159-b1e6-329c949d01e4	None	None		None
822e6305-b7a4-4df6-8f7b-74d117171381	icmp	0.0.0.0/0		None
c409c480-d296-4009-abf9-3e01c3b2c6ca	None	None		None

```
[ubuntu@workstation (keystone-admin)]:~$
```

1.21. Add another security rule to allow remote connection using SSH on the default port 22.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack security group rule create \
> --protocol tcp \
> --dst-port 22 \
> secgroup2
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack security group rule create \
> --protocol tcp \
> --dst-port 22 \
> secgroup2
```

Field	Value
created_at	2025-06-28T16:07:08Z
description	
direction	ingress
ether_type	IPv4
id	3af8f304-cd2c-4e3c-a68a-fa04e3155c94
name	None
port_range_max	22
port_range_min	22
project_id	39e851b14f864573aad60582c35e40dc
protocol	tcp
remote_group_id	None
remote_ip_prefix	0.0.0.0/0
revision_number	0
security_group_id	d4792861-920e-4dd6-bdd1-e9ea3b5f6d6a
updated_at	2025-06-28T16:07:08Z

```
[ubuntu@workstation (keystone-admin)]:~$
```

1.22. List the rules in the **secgroup2** security group again to ensure the SSH rule was created successfully.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack security group rule list \
> secgroup2
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack security group rule list \
> secgroup2
```

ID	IP Protocol	IP Range	Port Range	Remote Security Group
3af8f304-cd2c-4e3c-a68a-fa04e3155c94	tcp	0.0.0.0/0	22:22	None
4ca422bf-c170-4159-b1e6-329c949d01e4	None	None		None
822e6305-b7a4-4df6-8f7b-74d117171381	icmp	0.0.0.0/0		None
c409c480-d296-4009-abf9-3e01c3b2c6ca	None	None		None

```
[ubuntu@workstation (keystone-admin)]:~$
```

1.23. Leave the terminal window open and continue to the next task.

2 Applying Security Groups to Instances

In order for a security group's rules to apply to instance, the security group must be applied to one of the instance's interfaces. In these labs, instances will only have one interface, so a security group can be thought of as applying to the whole instance. Additionally, an interface may have more than one security group applied. Security groups can be added to an instance when the instance is created, and they can be added, removed, or modified at any time. This section will walk through adding and removing an instance's security groups with both the *OpenStack Unified CLI* and the *Horizon Dashboard*.

- 2.1. If a terminal window is not already open, open one and source the admin credentials from the `~/keystonerc-admin` file.

```
ubuntu@workstation:~$ source ~/keystonerc-admin
```

```
ubuntu@workstation:~$ source ~/keystonerc-admin
[ubuntu@workstation (keystone-admin)]:~$
```

- 2.2. A security group can be applied to an instance at creation from the command line. Create an instance with the security group **secgroup2** applied to it.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack server create \
> --image ubuntu \
> --flavor m1.small \
> --network shared \
> --security-group secgroup2 \
> instance1
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack server create \
> --image ubuntu \
> --flavor m1.small \
> --network shared \
> --security-group secgroup2 \
> instance1
```

Field	Value
OS-DCF:diskConfig	MANUAL
OS-EXT-AZ:availability_zone	
OS-EXT-SRV-ATTR:host	None
OS-EXT-SRV-ATTR:hypervisor_hostname	None
OS-EXT-SRV-ATTR:instance_name	
OS-EXT-STS:power_state	NOSTATE
OS-EXT-STS:task_state	scheduling
OS-EXT-STS:vm_state	building
OS-SRV-USG:launched_at	None
OS-SRV-USG:terminated_at	None
accessIPv4	
accessIPv6	
addresses	
adminPass	PTPvFbjoJb8A
config_drive	
created	2025-06-28T16:10:49Z
flavor	m1.small (2)
hostId	
id	f410ecba-01c0-47d6-b5ef-e991f54cff55
image	ubuntu (329d361e-f6dc-4b72-b200-3de0ec230e65)
key_name	None
name	instance1
progress	0
project_id	39e851b14f864573aad60582c35e40dc
properties	
security_groups	name='d4792861-920e-4dd6-bdd1-e9ea3b5f6d6a'
status	BUILD
updated	2025-06-28T16:10:49Z
user_id	14f5376f00c04e90b7103dd8d4263040
volumes_attached	

```
[ubuntu@workstation (keystone-admin)]:~$
```

- 2.3. The output of the previous step should have shown the ID of **secgroup2** in the **security_groups** row. To get a result that is easier to understand and verify that **secgroup2** is attached to the instance, we can show the details of the instance with a couple extra arguments.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack server show \  
> -c security_groups \  
> instance1
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack server show \  
> -c security_groups \  
> instance1  
+-----+-----+  
| Field          | Value          |  
+-----+-----+  
| security_groups | name='secgroup2' |  
+-----+-----+  
[ubuntu@workstation (keystone-admin)]:~$
```

Tip

The **-c security_groups** argument specifies that we want only the **security_groups** row in the output.

- 2.4. Remove **secgroup2** from the instance.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack server remove security group \  
> instance1 secgroup2
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack server remove security group \  
> instance1 \  
> secgroup2  
keys: ['name']  
[ubuntu@workstation (keystone-admin)]:~$
```

Note

This command may echo the string **keys: ['name']**. Ignore this output; the command still works as intended.

- 2.5. Verify that the security group is no longer applied to the instance. You should receive an error message saying the column name is not recognized.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack server show \
> -c security_groups \
> instance1
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack server show \
> -c security_groups \
> instance1
No recognized column names in [u'security_groups']. Recognized columns are (u'OS-D
CF:diskConfig', u'OS-EXT-AZ:availability_zone', u'OS-EXT-SRV-ATTR:host', u'OS-EXT-
SRV-ATTR:hypervisor_hostname', u'OS-EXT-SRV-ATTR:instance_name', u'OS-EXT-STS:powe
r_state', u'OS-EXT-STS:task_state', u'OS-EXT-STS:vm_state', u'OS-SRV-USG:launched_
at', u'OS-SRV-USG:terminated_at', u'accessIPv4', u'accessIPv6', u'addresses', u'co
nfig_drive', u'created', u'flavor', u'hostId', u'id', u'image', u'key_name', u'nam
e', u'progress', 'project_id', 'properties', u'status', u'updated', u'user_id', 'v
olumes_attached').
[ubuntu@workstation (keystone-admin)]:~$
```

- 2.6. A security group can also be added to an instance after the instance is created. Add **secgroup2** back to the instance.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack server add security group \
> instance1 secgroup2
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack server add security group \
> instance1 \
> secgroup2
keys: ['name']
[ubuntu@workstation (keystone-admin)]:~$
```

Note

This command may echo the string **keys: ['name']**. Ignore this output; the command still works as intended.

- 2.7. Verify that the security group is once again applied to the instance.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack server show \
> -c security_groups \
> instance1
```

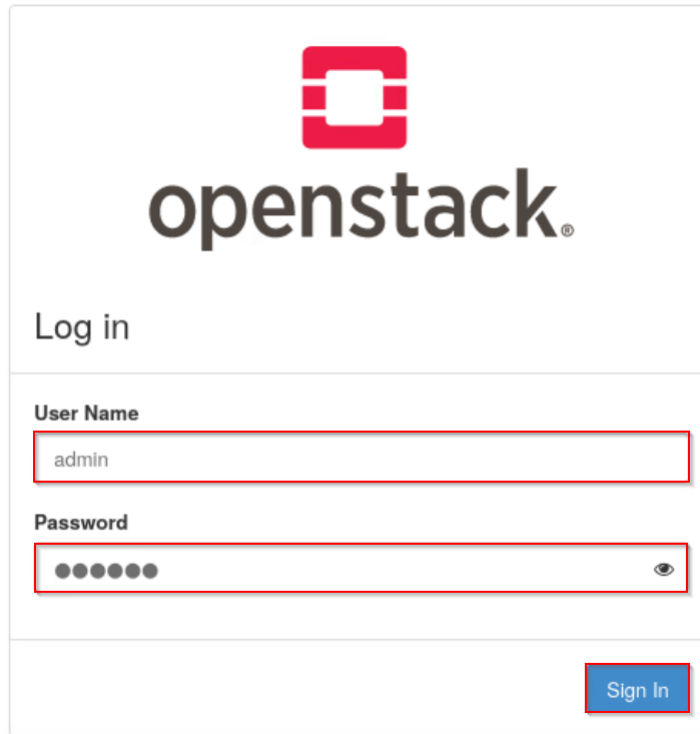
```
[ubuntu@workstation (keystone-admin)]:~$ openstack server show \
> -c security_groups \
> instance1
+-----+-----+
| Field          | Value          |
+-----+-----+
| security_groups | name='secgroup2' |
+-----+-----+
[ubuntu@workstation (keystone-admin)]:~$
```


- 2.8. Delete **instance1**. We will recreate it from the Horizon Dashboard to demonstrate adding a security group to an instance at creation time from the dashboard.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack server delete instance1
```

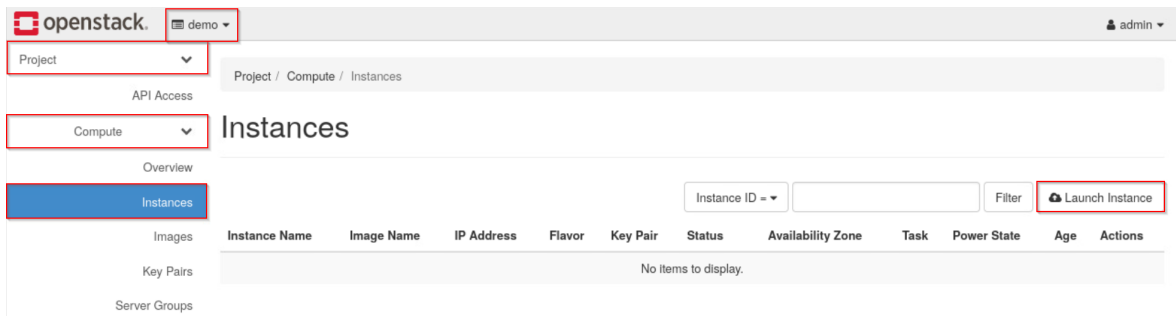
```
[ubuntu@workstation (keystone-admin)]:~$ openstack server delete instance1
[ubuntu@workstation (keystone-admin)]:~$
```

- 2.9. Leave the terminal window open, and open the web browser. Navigate to **192.168.1.20**, and log in to the dashboard as **admin** with the password **secret**.



The image shows the OpenStack login page. At the top is the OpenStack logo. Below it is the text "Log in". There are two input fields: "User Name" with the value "admin" and "Password" with masked characters. A "Sign In" button is at the bottom right.

- 2.10. Select the **demo** project. Navigate to **Project > Compute > Instances**, and click **Launch Instance**.



The image shows the OpenStack Horizon dashboard. The top navigation bar includes the OpenStack logo, a "demo" project selector, and a user profile "admin". The left sidebar has a "Project" dropdown, "API Access", "Compute" dropdown, "Overview", "Instances" (highlighted), "Images", "Key Pairs", and "Server Groups". The main content area shows the breadcrumb "Project / Compute / Instances" and the title "Instances". There is a search bar for "Instance ID", a "Filter" button, and a "Launch Instance" button. Below is a table with columns: Instance Name, Image Name, IP Address, Flavor, Key Pair, Status, Availability Zone, Task, Power State, Age, and Actions. The table currently shows "No items to display."

2.11. In the *Details* tab, type **instance1** in the *Instance Name* field. Click **Next**.

Launch Instance ✕

Details

Source *

Flavor *

Networks *

Network Ports

Security Groups

Key Pair

Configuration

Server Groups

Scheduler Hints

Metadata

Please provide the initial hostname for the instance, the availability zone where it will be deployed, and the instance count. Increase the Count to create multiple instances with the same settings.

Project Name

Instance Name *

Description

Availability Zone

Count *

Total Instances (10 Max)

10%

0 Current Usage
1 Added
9 Remaining

✕ Cancel

< Back

Next >

Launch Instance

- 2.12. In the *Source* tab, set *Create New Volume* to **No**, and scroll down (if needed) to select the **ubuntu** image. Click **Next**.

Launch Instance

Details

Source

Flavor

Networks

Network Ports

Security Groups

Key Pair

Configuration

Server Groups

Scheduler Hints

Metadata

Instance source is the template used to create an instance. You can use an image, a snapshot of an instance (image snapshot), a volume or a volume snapshot (if enabled). You can also choose to use persistent storage by creating a new volume.

Select Boot Source

Image

Create New Volume

Yes No

Allocated

Displaying 0 items

Name	Updated	Size	Format	Visibility
Select an item from Available items below				

Displaying 0 items

Available 2

Select one

Click here for filters or full text search.

Displaying 2 items

Name	Updated	Size	Format	Visibility
> cirros-0.6.2-x86_64-disk	2/9/24 7:59 PM	20.44 MB	QCOW2	Public
> ubuntu	2/9/24 9:32 PM	647.50 MB	QCOW2	Shared

Displaying 2 items

Cancel

< Back Next > Launch Instance

Stop

Before proceeding to the next step, confirm that **ubuntu** appears in the *Allocated* section.

2.13. In the *Flavor* tab, scroll down (if needed) to select the **m1.small** flavor. Click **Next**.

Launch Instance

Details

Source

Flavor

Networks

Network Ports

Security Groups

Key Pair

Configuration

Server Groups

Scheduler Hints

Metadata

Flavors manage the sizing for the compute, memory and storage capacity of the instance.

Allocated

Displaying 0 Items

Name	VCPUS	RAM	Total Disk	Root Disk	Ephemeral Disk	Public
Select a flavor from the available flavors below.						

Displaying 0 Items

▼ Available 12

Select one

Click here for filters or full text search.

Displaying 12 Items

Name	VCPUS	RAM	Total Disk	Root Disk	Ephemeral Disk	Public
m1.nano	1	128 MB	1 GB	1 GB	0 GB	Yes
m1.micro	1	192 MB	1 GB	1 GB	0 GB	Yes
cirros256	1	256 MB	1 GB	1 GB	0 GB	Yes
m1.tiny	1	512 MB	1 GB	1 GB	0 GB	Yes
ds512M	1	512 MB	5 GB	5 GB	0 GB	Yes
ds1G	1	1 GB	10 GB	10 GB	0 GB	Yes
m1.small	1	2 GB	20 GB	20 GB	0 GB	Yes
ds2G	2	2 GB	10 GB	10 GB	0 GB	Yes
m1.medium	2	4 GB	40 GB	40 GB	0 GB	Yes
ds4G	4	4 GB	20 GB	20 GB	0 GB	Yes
m1.large	4	8 GB	80 GB	80 GB	0 GB	Yes
m1.xlarge	8	16 GB	160 GB	160 GB	0 GB	Yes

Displaying 12 Items

Cancel

< Back

Next >

Launch Instance

Stop

Before proceeding to the next step, confirm that **m1.small** appears in the *Allocated* section.

2.14. In the *Networks* tab, select the **shared** network. Navigate to the **Security Groups** tab.

Launch Instance ✕

[Details](#)
[Source](#)
[Flavor](#)
[Networks](#)
[Network Ports](#)
[Security Groups](#)
[Key Pair](#)
[Configuration](#)
[Server Groups](#)
[Scheduler Hints](#)
[Metadata](#)

Networks provide the communication channels for instances in the cloud. You can select ports instead of networks or a mix of both.

▼ **Allocated**
Displaying 0 items

Network	Subnets Associated	Shared	Admin State	Status
Select one or more networks from the available networks below.				

Displaying 0 items

▼ **Available** 2 Select one or more

Click here for filters or full text search. ✕

Displaying 2 items

Network	Subnets Associated	Shared	Admin State	Status
➤ shared	shared-subnet	Yes	Up	Active ⬆
➤ private	ipv6-private-subnet private-subnet	No	Up	Active ⬆

Displaying 2 items

✕ Cancel < Back Next > Launch Instance

Stop

Before proceeding to the next step, confirm that **shared** appears in the *Allocated* section.

- 2.15. In the *Security Groups* tab, deselect the **default** security group since we have our own to apply. Scroll down (if needed) to select the **secgroup2** security group. This will apply the security group to all interfaces on the instance. Click **Launch Instance**.

Launch Instance ✕

Details

Source

Flavor

Networks

Network Ports

Security Groups

Key Pair

Configuration

Server Groups

Scheduler Hints

Metadata

Select the security groups to launch the instance in.

▼ Allocated 1

Displaying 1 item

Name	Description
> default	Default security group

Displaying 1 item

▼ Available 1

Select one or more

Click here for filters or full text search.

Displaying 1 item

Name	Description
> secgroup2	secgroup2

Displaying 1 item

✕ Cancel

< Back

Next >

Launch Instance

Stop

Before proceeding to the next step, confirm that **secgroup2** appears in the *Allocated* section.

- 2.16. You should be redirected to the **Project > Compute > Instances** page. The security groups and rules applied to an instance can be found by clicking the instance's name.

openstack demo admin

Project / Compute / Instances

Instances

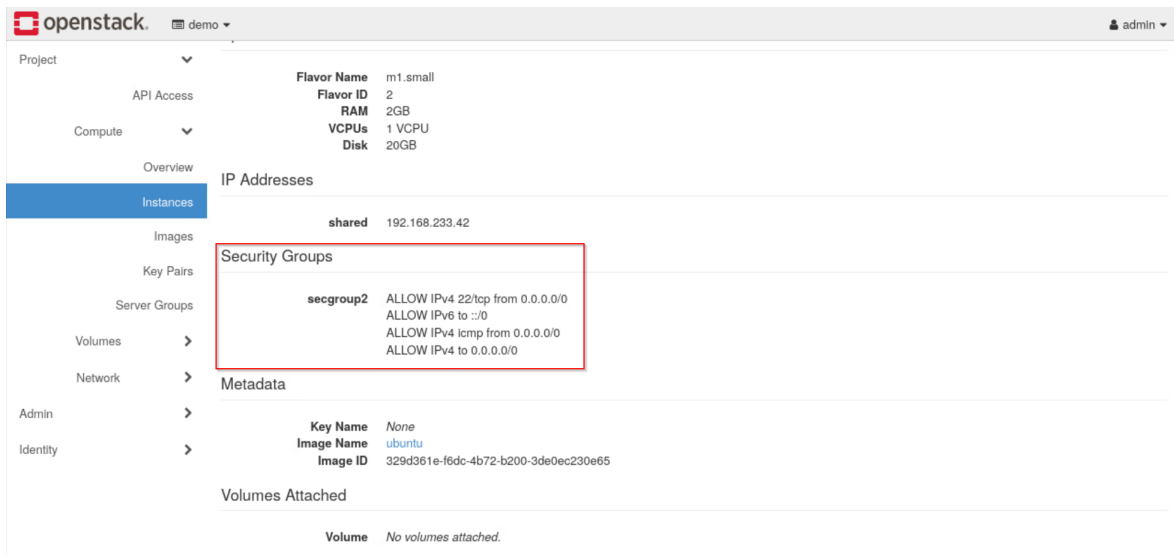
Instance ID ▾ Filter Launch Instance Delete Instances More Actions

Displaying 1 item

Instance Name	Image Name	IP Address	Flavor	Key Pair	Status	Availability Zone	Task	Power State	Age	Actions	
Instance1	ubuntu	192.168.233.42	m1.small	-	Active	us-east-1a	nova	None	Running	1 minute	Create Snapshot

Displaying 1 item

- 2.17. On the *Overview* tab of the *instance1* page, scroll down to view the *Security Groups* section. In this case, you should see that **secgroup2** allows all IPv4 and IPv6 egress traffic by default, and you should see the two rules we added for ICMP and SSH (TCP port 22).



openstack demo admin

Project

API Access

Compute

Overview

Instances

Images

Key Pairs

Server Groups

Volumes

Network

Admin

Identity

Flavor Name: m1.small

Flavor ID: 2

RAM: 2GB

VCPUs: 1 VCPU

Disk: 20GB

IP Addresses

shared 192.168.233.42

Security Groups

Security Group	Rules
secgroup2	ALLOW IPv4 22/tcp from 0.0.0.0/0 ALLOW IPv6 to ::0 ALLOW IPv4 icmp from 0.0.0.0/0 ALLOW IPv4 to 0.0.0.0/0

Metadata

Key Name: None

Image Name: ubuntu

Image ID: 329d361e-f6dc-4b72-b200-3de0ec230e65

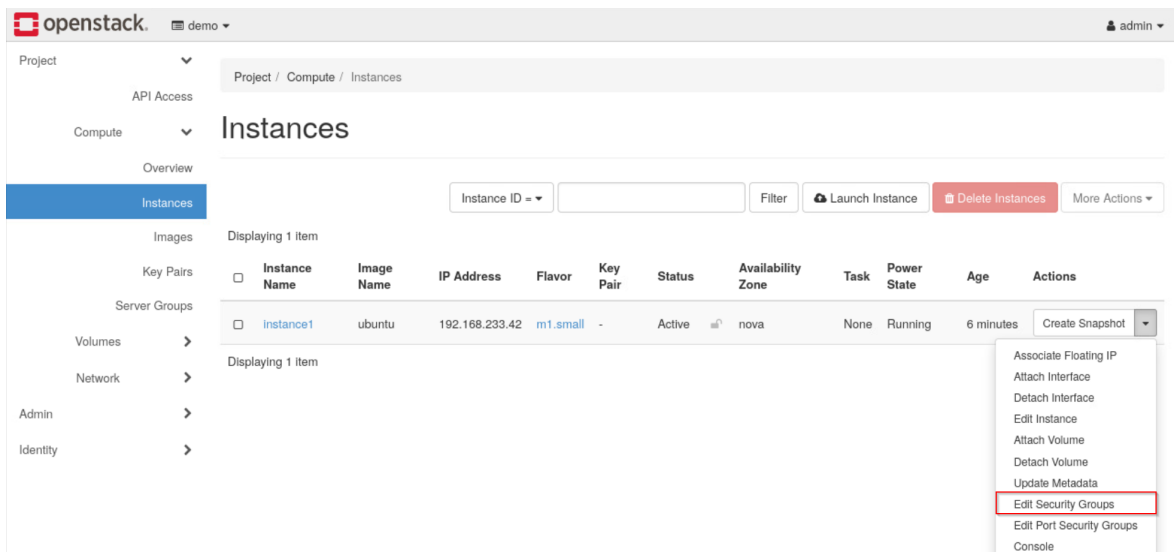
Volumes Attached

Volume: No volumes attached.

Note

If no security group is applied to an instance, the *Security Groups* section will simply say "Not available".

- 2.18. Navigate back to **Project > Compute > Instances**. To view and edit the security groups of an instance from the dashboard, click the dropdown next to **Create Snapshot**, and click **Edit Security Groups**.



openstack demo admin

Project / Compute / Instances

Instances

Instance ID: [dropdown] Filter Launch Instance Delete Instances More Actions

Displaying 1 item

Instance Name	Image Name	IP Address	Flavor	Key Pair	Status	Availability Zone	Task	Power State	Age	Actions
Instance1	ubuntu	192.168.233.42	m1.small	-	Active	nova	None	Running	6 minutes	Create Snapshot Associate Floating IP Attach Interface Detach Interface Edit Instance Attach Volume Detach Volume Update Metadata Edit Security Groups Edit Port Security Groups Console

Displaying 1 item

Note

This option will edit security groups for all interfaces on the instance. Notice that there is also an option to **Edit Port Security Groups**, which allows you to edit security groups for individual interfaces on the instance.

- 2.19. In the **Edit Instance** popup, the **Instance Security Groups** list contains the security groups that are currently applied to the instance. Remove **secgroup2** from this instance by clicking the - button next to that group, and click **Save** to finalize the change.

Edit Instance

Information *Security Groups

Add and remove security groups to this instance from the list of available security groups.

Warning: If you change security groups here, the change will be applied to all interfaces of the instance. If you have multiple interfaces on this instance and apply different security groups per port, use "Edit Port Security Groups" action instead.

All Security Groups

Filter

default +

Instance Security Groups

Filter

secgroup2 -

CancelSave

- 2.20. Re-open the popup by clicking the dropdown next to **Create Snapshot** and clicking **Edit Security Groups**. The **All Security Groups** list contains the security groups that are available but are not currently applied to the instance. Add the **default** security group to the interface by clicking the **+** button next to that group, and click **Update** to finalize the change.

Edit Instance ✕

Information * Security Groups

Add and remove security groups to this instance from the list of available security groups.

Warning: If you change security groups here, the change will be applied to all interfaces of the instance. If you have multiple interfaces on this instance and apply different security groups per port, use "Edit Port Security Groups" action instead.

All Security Groups

Filter

default	+
secgroup2	+

Instance Security Groups

Filter

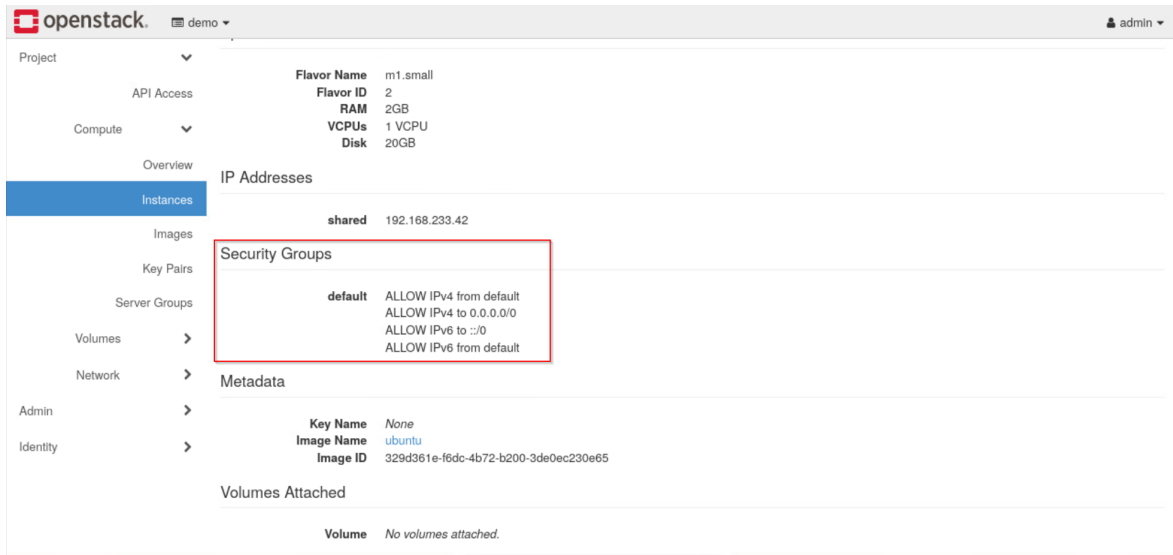
No security groups enabled.

Cancel Save

Tip

You can assign and unassign multiple security groups before saving the changes.

- 2.21. To verify that the security group has been applied again, click **instance1**. On the *Overview* tab of the *instance1* page, scroll down to view the *Security Groups* section. This time, you should see the **default** security group and its rules listed.



The screenshot shows the OpenStack dashboard for a project named 'demo'. The left sidebar contains navigation links: Project, API Access, Compute, Overview, Instances (selected), Images, Key Pairs, Server Groups, Volumes, Network, Admin, and Identity. The main content area shows details for 'instance1'. The 'Security Groups' section is highlighted with a red box, displaying the 'default' security group with the following rules:

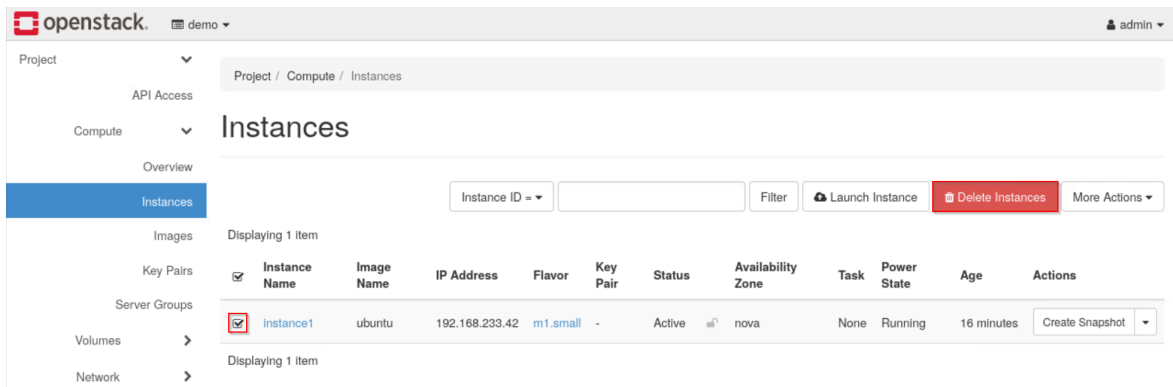
Security Group	Rules
default	- ALLOW IPv4 from default - ALLOW IPv4 to 0.0.0.0/0 - ALLOW IPv6 to ::0 - ALLOW IPv6 from default

Below the Security Groups section, the 'Metadata' section shows:

Key Name	Value
Key Name	None
Image Name	ubuntu
Image ID	329d361e-f6dc-4b72-b200-3de0ec230e65

The 'Volumes Attached' section shows 'No volumes attached.'

- 2.22. We no longer need this instance, so navigate back to **Project > Compute > Instances**, select **instance1**, and click **Delete Instances**.



The screenshot shows the OpenStack dashboard for a project named 'demo'. The left sidebar contains navigation links: Project, API Access, Compute, Overview, Instances (selected), Images, Key Pairs, Server Groups, Volumes, Network, Admin, and Identity. The main content area shows the 'Instances' list. The 'instance1' is selected, and the 'Delete Instances' button is visible.

Instance Name	Image Name	IP Address	Flavor	Key Pair	Status	Availability Zone	Task	Power State	Age	Actions
instance1	ubuntu	192.168.233.42	m1.small	-	Active	nova	None	Running	16 minutes	Create Snapshot

2.23. We will also recreate the security group in the final section of the lab in order to show a complete example, so we can safely delete **secgroup2**. To delete the security group from the dashboard, navigate to **Project > Network > Security Groups**. Click the dropdown next to the **Manage Rules** button in the same row as **secgroup2**, and then click **Delete Security Group**.

The screenshot shows the OpenStack dashboard interface. On the left sidebar, the 'Project' and 'Network' menu items are highlighted with red boxes. The main content area displays the 'Security Groups' page. At the top, there is a breadcrumb 'Project / Network / Security Groups' and a title 'Security Groups'. Below the title, there are buttons for '+ Create Security Group' and 'Delete Security Groups'. A table lists the security groups:

Name	Security Group ID	Description	Shared	Actions
default	62dafb67-e7fd-44d6-bf87-f4701dd66875	Default security group	False	Manage Rules
secgroup2	d4792861-920e-4dd6-bdd1-e9ea3b5f6d6a	secgroup2	False	Manage Rules (dropdown open)

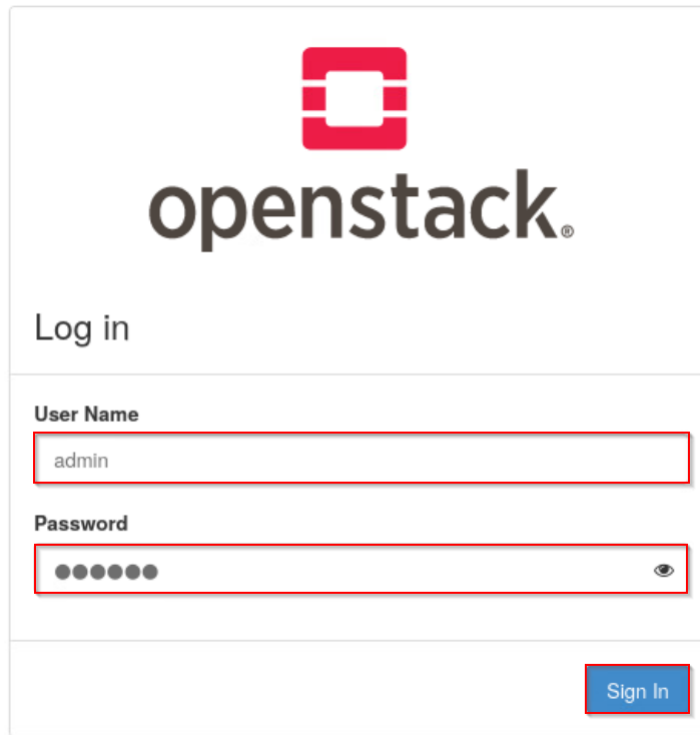
The dropdown menu for 'secgroup2' is open, showing 'Edit Security Group' and 'Delete Security Group' options, with 'Delete Security Group' highlighted by a red box.

2.24. Leave the web browser open, and continue to the next task.

3 Creating SSH Key Pairs

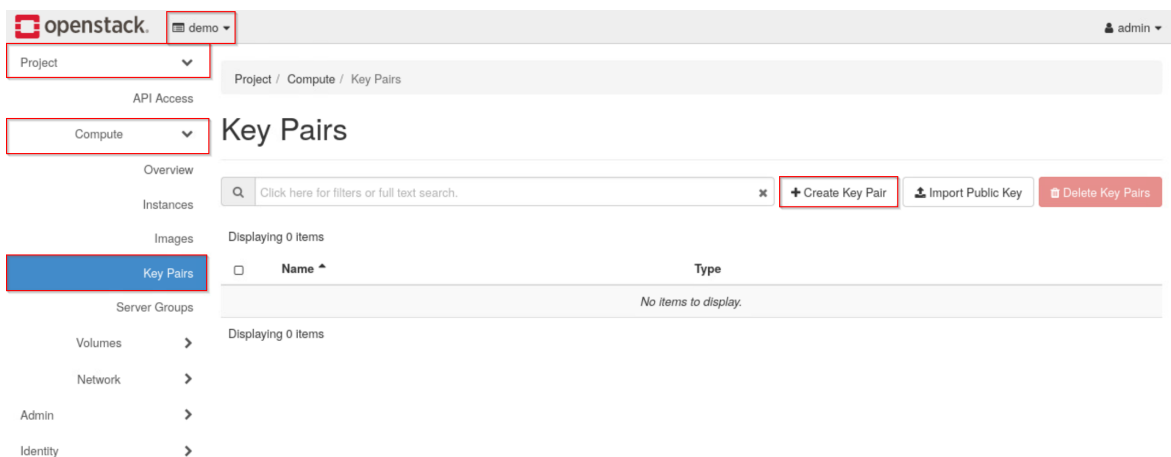
In this task, you will use the *Horizon Dashboard* and *OpenStack Unified CLI* to create and manage SSH key pairs. These keys will later be used to securely connect to external instances from outside the OpenStack environment.

- 3.1. If the web browser is not already open, open it. Navigate to **192.168.1.20**, and log in to the dashboard as **admin** with the password **secret**.



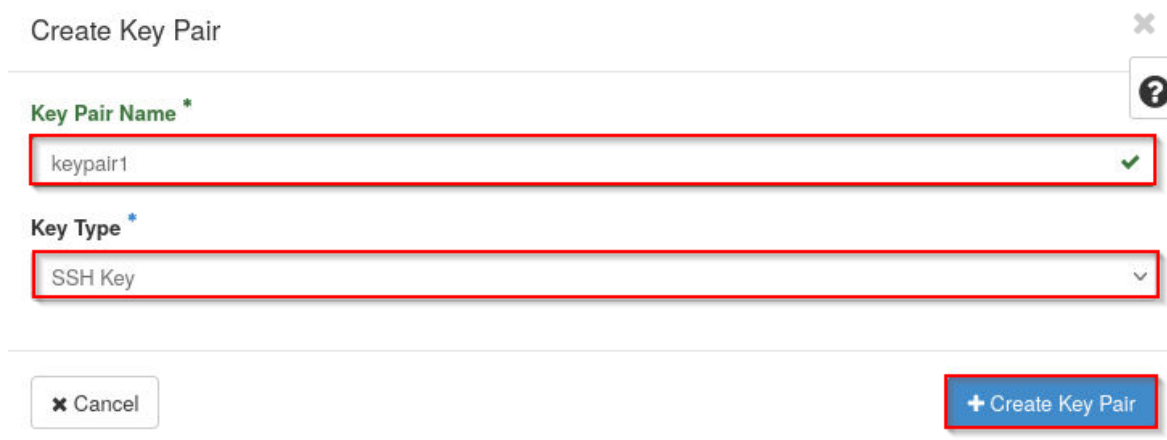
The image shows the OpenStack login page. At the top is the OpenStack logo. Below it is the text "Log in". There are two input fields: "User Name" with the value "admin" and "Password" with masked characters. A "Sign In" button is at the bottom right.

- 3.2. Select the **demo** project, and navigate to **Project > Compute > Key Pairs**. Click **Create Key Pair** to create a new key pair.



The image shows the OpenStack Horizon dashboard. The top bar includes the OpenStack logo, a "demo" project selector, and a user profile "admin". The left sidebar has a navigation menu with "Project", "Compute", "Key Pairs", "Volumes", "Network", "Admin", and "Identity". The main content area shows the "Key Pairs" page for the "demo" project. It includes a search bar, a "+ Create Key Pair" button, and a table with columns "Name" and "Type". The table is currently empty, displaying "No items to display."

- 3.3. Enter **keypair1** in the *Key Pair Name* field, and select **SSH Key** in the *Key Type* dropdown. Click **Create Key Pair**. This will create the key pair and download it to the `~/Downloads` directory.



Create Key Pair

Key Pair Name *

keypair1

Key Type *

SSH Key

Cancel Create Key Pair

Tip

When creating key pairs from the Horizon Dashboard, the private key file permissions are *not* set strictly, so it is recommended to still set them through the command line. We will do this in the following steps.

- 3.4. Sign out of the Horizon Dashboard and close the web browser.
- 3.5. If a terminal window is not already open, open one and source the **admin** credentials from the `~/keystonerc-admin` file.

```
ubuntu@workstation:~$ source ~/keystonerc-admin
```

```
ubuntu@workstation:~$ source ~/keystonerc-admin
[ubuntu@workstation (keystone-admin)]:~$
```

- 3.6. To better protect the private key, use the **chmod** command with a mode of **600** to make it so that the **ubuntu** user has read/write permissions on the private key file, and groups and other users have no permissions to the file.

```
[ubuntu@workstation (keystone-admin)]:~$ chmod 600 ~/Downloads/keypair1.pem
```

```
[ubuntu@workstation (keystone-admin)]:~$ chmod 600 ~/Downloads/keypair1.pem
[ubuntu@workstation (keystone-admin)]:~$
```

- 3.7. We only need one key pair to connect to the external instance, so the key pair created from the Horizon Dashboard can safely be deleted in order to demonstrate creating a key pair from the command line. Delete the **keypair1** key pair.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack keypair delete keypair1
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack keypair delete keypair1
[ubuntu@workstation (keystone-admin)]:~$
```

Note

Note that a key pair in the context of OpenStack is actually a misnomer. The key pair object really only refers to the public key, while the private key only exists in the file in which it is saved. Therefore, the private key file will still exist after deleting the key pair.

- 3.8. Delete the private key located at **~/Downloads/keypair1.pem**.

```
[ubuntu@workstation (keystone-admin)]:~$ rm -f ~/Downloads/keypair1.pem
```

```
[ubuntu@workstation (keystone-admin)]:~$ rm -f ~/Downloads/keypair1.pem
[ubuntu@workstation (keystone-admin)]:~$
```

- 3.9. List the available key pairs to verify that **keypair1** was deleted.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack keypair list
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack keypair list
[ubuntu@workstation (keystone-admin)]:~$
```

- 3.10. Create the key pair **keypair2**, and save the private key to the file **~/Downloads/keypair2.pem**.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack keypair create \
> keypair2 > ~/Downloads/keypair2.pem
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack keypair create \
> keypair2 > ~/Downloads/keypair2.pem
[ubuntu@workstation (keystone-admin)]:~$
```

- 3.11. Set the file permissions of the private key so that only the **ubuntu** user has read/write permissions.

```
[ubuntu@workstation (keystone-admin)]:~$ chmod 600 ~/Downloads/keypair2.pem
```

```
[ubuntu@workstation (keystone-admin)]:~$ chmod 600 ~/Downloads/keypair2.pem
[ubuntu@workstation (keystone-admin)]:~$
```

3.12. List the available key pairs to verify the creation of **keypair2**.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack keypair list
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack keypair list
+-----+-----+
| Name      | Fingerprint |
+-----+-----+
| keypair2  | 6f:46:7c:56:38:82:f3:d2:f6:97:b1:65:ae:c0:90:87 |
+-----+-----+
[ubuntu@workstation (keystone-admin)]:~$
```

3.13. Leave the terminal window open and continue to the next task.

4 Applying SSH Keys to Instances

In this task, you will use both the *Horizon Dashboard* and *OpenStack Unified CLI* to apply SSH key pairs to instances. This is the final piece necessary to have an instance that can be reached and logged in to from an external network, which the next task will explore.

- 4.1. If a terminal window is not already open, open one and source the admin credentials from the `~/keystonerc-admin` file.

```
ubuntu@workstation:~$ source ~/keystonerc-admin
```

```
ubuntu@workstation:~$ source ~/keystonerc-admin  
[ubuntu@workstation (keystone-admin)]:~$
```


- 4.2. An SSH key can be applied to an instance at creation from the command line. Create an instance with the **keypair2** key pair.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack server create \
> --image ubuntu \
> --flavor m1.small \
> --network shared \
> --key-name keypair2 \
> instance1
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack server create \
> --image ubuntu \
> --flavor m1.small \
> --network shared \
> --key-name keypair2 \
> instance1
```

Field	Value
OS-DCF:diskConfig	MANUAL
OS-EXT-AZ:availability_zone	
OS-EXT-SRV-ATTR:host	None
OS-EXT-SRV-ATTR:hypervisor_hostname	None
OS-EXT-SRV-ATTR:instance_name	
OS-EXT-STS:power_state	NOSTATE
OS-EXT-STS:task_state	scheduling
OS-EXT-STS:vm_state	building
OS-SRV-USG:launched_at	None
OS-SRV-USG:terminated_at	None
accessIPv4	
accessIPv6	
addresses	
adminPass	XTL7K7hgoZwi
config_drive	
created	2025-06-30T15:52:26Z
flavor	m1.small (2)
hostId	
id	a498cf56-8910-4525-b446-a48141b6b7b7
image	ubuntu (329d361e-f6dc-4b72-b200-3de0ec230e65)
key_name	keypair2
name	instance1
progress	0
project_id	39e851b14f864573aad60582c35e40dc
properties	
security_groups	name='default'
status	BUILD
updated	2025-06-30T15:52:26Z
user_id	14f5376f00c04e90b7103dd8d4263040
volumes_attached	

```
[ubuntu@workstation (keystone-admin)]:~$
```

Tip

OpenStack does not provide a way to directly change the key pair associated with an existing instance. We will not need to in these labs, but if you need to use a different SSH key, one option is to manually add your new key to the `/.ssh/authorized_keys` file inside the instance. However, this change will not be tracked by OpenStack. Another option is to rebuild the instance with a new key pair. If the instance contains important data, it's important to back it up—for example, with a snapshot (covered later).

4.3. Verify that `keypair2` is attached to `instance1`.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack server show \  
> -c key_name \  
> instance1
```

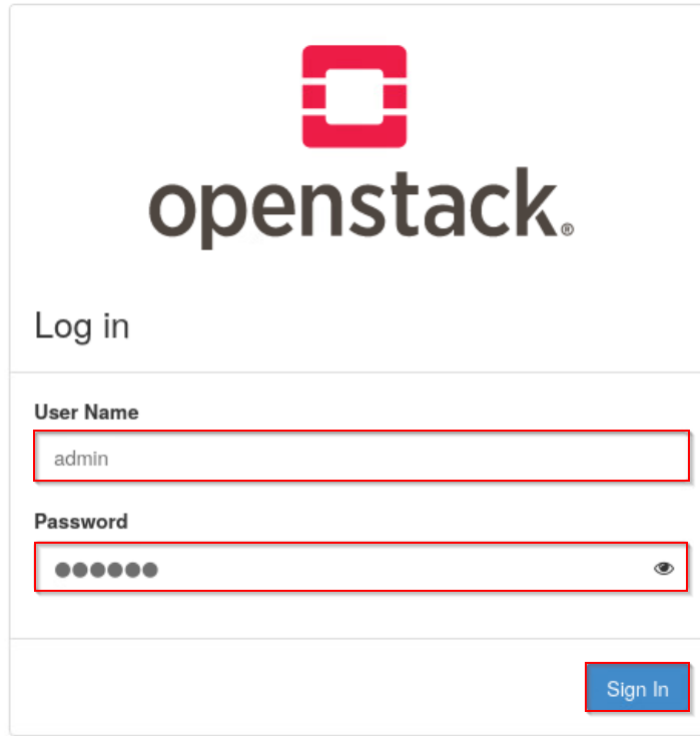
```
[ubuntu@workstation (keystone-admin)]:~$ openstack server show \  
> -c key_name \  
> instance1  
+-----+-----+  
| Field  | Value  |  
+-----+-----+  
| key_name | keypair2 |  
+-----+-----+  
[ubuntu@workstation (keystone-admin)]:~$
```

4.4. Delete `instance1`. We will recreate it from the Horizon Dashboard to demonstrate adding a security group to an instance at creation time from the dashboard.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack server delete instance1
```

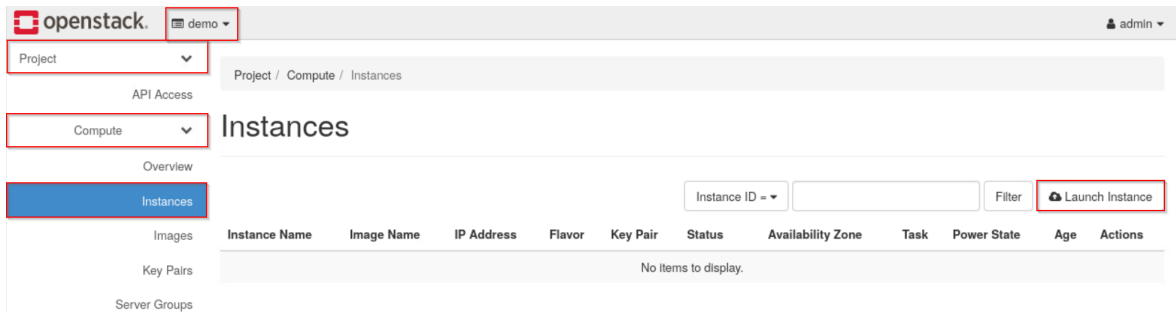
```
[ubuntu@workstation (keystone-admin)]:~$ openstack server delete instance1  
[ubuntu@workstation (keystone-admin)]:~$
```

- 4.5. Leave the terminal window open, and open the web browser. Navigate to **192.168.1.20**, and log in to the dashboard as **admin** with the password **secret**.



The image shows the OpenStack login page. At the top is the OpenStack logo. Below it is the text "Log in". There are two input fields: "User Name" with the value "admin" and "Password" with masked characters. A "Sign In" button is at the bottom right.

- 4.6. Select the **demo** project. Navigate to **Project > Compute > Instances**, and click **Launch Instance**.



The image shows the OpenStack dashboard. The top bar has the OpenStack logo, a "demo" project selector, and a user dropdown showing "admin". The left sidebar has a "Project" dropdown, "API Access", "Compute" dropdown, "Overview", "Instances" (highlighted), "Images", "Key Pairs", and "Server Groups". The main content area shows the breadcrumb "Project / Compute / Instances" and the title "Instances". There is a "Launch Instance" button in the top right. Below the title is a table with columns: Instance Name, Image Name, IP Address, Flavor, Key Pair, Status, Availability Zone, Task, Power State, Age, and Actions. The table is currently empty with the message "No items to display."

4.7. In the *Details* tab, type **instance1** in the *Instance Name* field. Click **Next**.

Launch Instance ✕

Details

Source *

Flavor *

Networks *

Network Ports

Security Groups

Key Pair

Configuration

Server Groups

Scheduler Hints

Metadata

Please provide the initial hostname for the instance, the availability zone where it will be deployed, and the instance count. Increase the Count to create multiple instances with the same settings.

Project Name

demo

Instance Name *

instance1

Description

Availability Zone

nova

Count *

1

Total Instances (10 Max)

10%

0 Current Usage
1 Added
9 Remaining

✕ Cancel < Back **Next >** Launch Instance

- 4.8. In the *Source* tab, set *Create New Volume* to **No**, and scroll down (if needed) to select the **ubuntu** image. Click **Next**.

Launch Instance

Details

Source

Flavor

Networks

Network Ports

Security Groups

Key Pair

Configuration

Server Groups

Scheduler Hints

Metadata

Instance source is the template used to create an instance. You can use an image, a snapshot of an instance (image snapshot), a volume or a volume snapshot (if enabled). You can also choose to use persistent storage by creating a new volume.

Select Boot Source

Image

Create New Volume

Yes No

Allocated

Displaying 0 items

Name	Updated	Size	Format	Visibility
Select an item from Available items below				

Displaying 0 items

Available 2

Select one

Click here for filters or full text search.

Displaying 2 items

Name	Updated	Size	Format	Visibility
> cirros-0.6.2-x86_64-disk	2/9/24 7:59 PM	20.44 MB	QCOW2	Public
> ubuntu	2/9/24 9:32 PM	647.50 MB	QCOW2	Shared

Displaying 2 items

Cancel

< Back

Next >

Launch Instance

Stop

Before proceeding to the next step, confirm that **ubuntu** appears in the *Allocated* section.

4.9. In the *Flavor* tab, scroll down (if needed) to select the **m1.small** flavor. Click **Next**.

Launch Instance

Details

Source

Flavor

Networks

Network Ports

Security Groups

Key Pair

Configuration

Server Groups

Scheduler Hints

Metadata

Flavors manage the sizing for the compute, memory and storage capacity of the instance.

Allocated

Displaying 0 Items

Name	VCPUS	RAM	Total Disk	Root Disk	Ephemeral Disk	Public
Select a flavor from the available flavors below.						

Displaying 0 Items

▼ Available 12

Select one

Click here for filters or full text search.

Displaying 12 Items

Name	VCPUS	RAM	Total Disk	Root Disk	Ephemeral Disk	Public
m1.nano	1	128 MB	1 GB	1 GB	0 GB	Yes
m1.micro	1	192 MB	1 GB	1 GB	0 GB	Yes
cirros256	1	256 MB	1 GB	1 GB	0 GB	Yes
m1.tiny	1	512 MB	1 GB	1 GB	0 GB	Yes
ds512M	1	512 MB	5 GB	5 GB	0 GB	Yes
ds1G	1	1 GB	10 GB	10 GB	0 GB	Yes
m1.small	1	2 GB	20 GB	20 GB	0 GB	Yes
ds2G	2	2 GB	10 GB	10 GB	0 GB	Yes
m1.medium	2	4 GB	40 GB	40 GB	0 GB	Yes
ds4G	4	4 GB	20 GB	20 GB	0 GB	Yes
m1.large	4	8 GB	80 GB	80 GB	0 GB	Yes
m1.xlarge	8	16 GB	160 GB	160 GB	0 GB	Yes

Displaying 12 Items

Cancel

< Back

Next >

Launch Instance

Stop

Before proceeding to the next step, confirm that **m1.small** appears in the *Allocated* section.

4.10. In the *Networks* tab, select the **shared** network. Navigate to the *Key Pair* tab.

Launch Instance

Details

Source

Flavor

Networks

Network Ports

Security Groups

Key Pair

Configuration

Server Groups

Scheduler Hints

Metadata

Networks provide the communication channels for instances in the cloud. You can select ports instead of networks or a mix of both.

▼ Allocated

Displaying 0 items

Network	Subnets Associated	Shared	Admin State	Status
Select one or more networks from the available networks below.				

Displaying 0 items

▼ Available 2

Select one or more

Click here for filters or full text search.

Displaying 2 items

Network	Subnets Associated	Shared	Admin State	Status
> shared	shared-subnet	Yes	Up	Active
> private	ipv6-private-subnet private-subnet	No	Up	Active

Displaying 2 items

Cancel

< Back

Next >

Launch Instance

Stop

Before proceeding to the next step, confirm that **shared** appears in the *Allocated* section.

- 4.11. In the *Key Pair* tab, select **keypair2** if it is not already in the *Allocated* section, and click **Launch Instance**.

Launch Instance ✕

Details

A key pair allows you to SSH into your newly created instance. You may select an existing key pair, import a key pair, or generate a new key pair.

[+ Create Key Pair](#) [Import Key Pair](#)

Allocated

Displaying 1 Item

Name	Type
keypair2	ssh

Displaying 1 Item

Available 0 Select one

✕

Displaying 0 Items

Name	Type
No items to display.	

Displaying 0 Items

✕ Cancel < Back Next > Launch Instance

Stop

Before proceeding to the next step, confirm that **keypair2** appears in the *Allocated* section.

- 4.12. You should be redirected to the **Project > Compute > Instances** page. To verify that the key pair was applied, first click **instance1** to go to the instance's details page.

openstack demo admin

Project / Compute / Instances

Instances

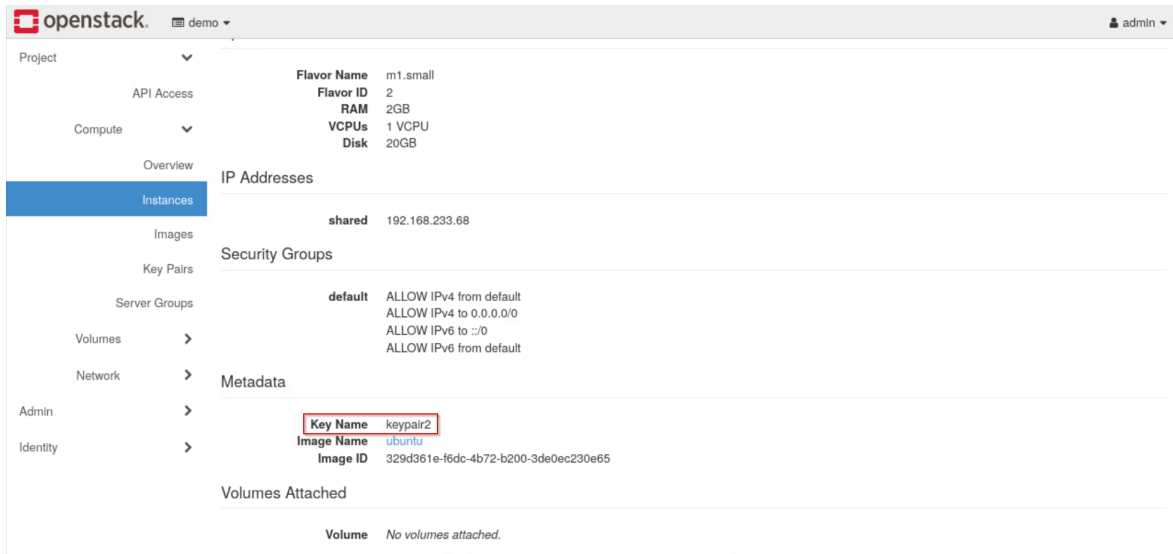
Instance ID Filter [Launch Instance](#) [Delete Instances](#) [More Actions](#)

Displaying 1 Item

Instance Name	Image Name	IP Address	Flavor	Key Pair	Status	Availability Zone	Task	Power State	Age	Actions
instance1	ubuntu	192.168.233.68	m1.small	keypair2	Active	nova	None	Running	4 minutes	Create Snapshot

Displaying 1 Item

- 4.13. Scroll down to the *Metadata* section and verify that **keypair2** is present in the *Key Name* row.

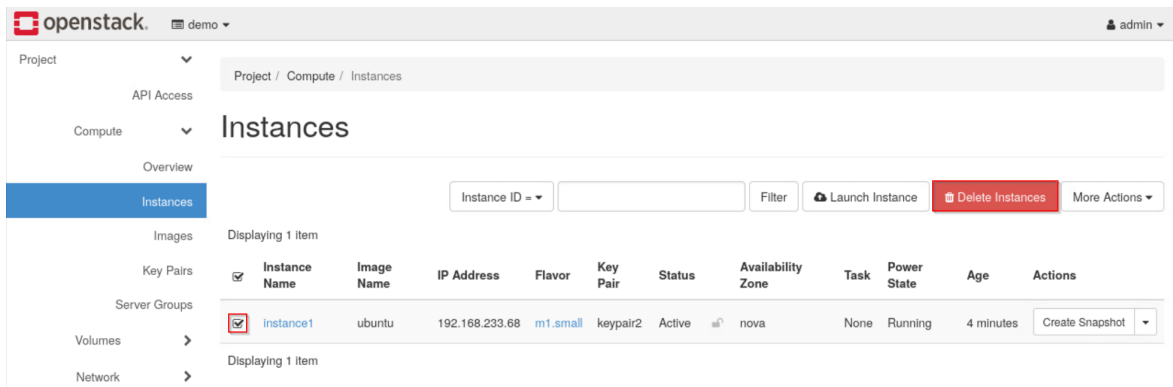


The screenshot shows the OpenStack dashboard interface. On the left, a sidebar contains navigation links: Project, API Access, Compute, Overview, Instances (selected), Images, Key Pairs, Server Groups, Volumes, Network, Admin, and Identity. The main content area displays instance details for 'demo'. Under the 'Metadata' section, the 'Key Name' row is highlighted with a red box, showing the value 'keypair2'. Other metadata rows include 'Image Name' (ubuntu) and 'Image ID' (329d361e-f6dc-4b72-b200-3de0ec230e65). The 'Volumes Attached' section shows 'No volumes attached'.

Note

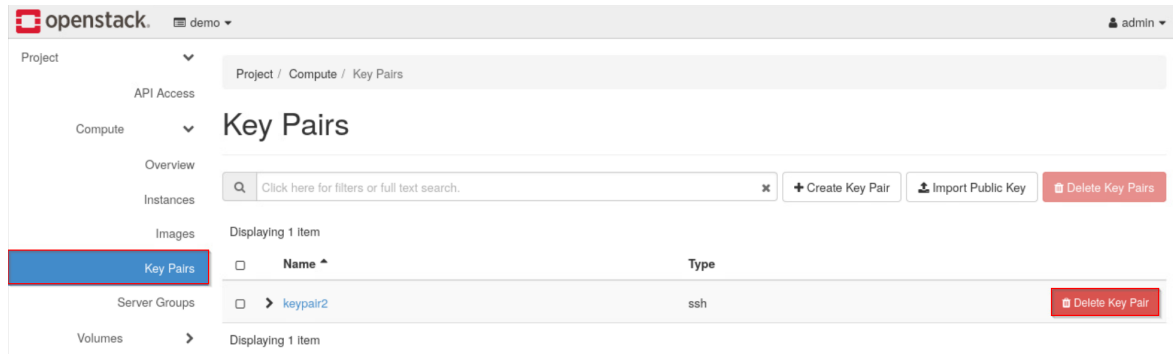
If an instance does not have a key pair, the *Key Name* row will show "None".

- 4.14. We no longer need this instance, so navigate back to **Project > Compute > Instances**, select **instance1**, and click **Delete Instances**.



The screenshot shows the OpenStack dashboard interface. The left sidebar is the same as in the previous screenshot. The main content area displays the 'Instances' list. At the top, there is a breadcrumb 'Project / Compute / Instances' and a table with columns: Instance ID, Filter, Launch Instance, Delete Instances, and More Actions. Below the table, there is a section titled 'Displaying 1 item' with a table of instance details. The table has columns: Instance Name, Image Name, IP Address, Flavor, Key Pair, Status, Availability Zone, Task, Power State, Age, and Actions. The first row, 'instance1', is selected with a red checkbox. The 'Delete Instances' button is highlighted in red.

- 4.15.** We will also recreate the key pair in the final section of the lab in order to show a complete example. So, we can safely delete **keypair2**. To delete the key pair from the dashboard, navigate to **Project > Compute > Key Pairs**, and click the **Delete Key Pair** button in the same row as **keypair2**.



- 4.16.** Log out of the *Horizon Dashboard*, and close the web browser. Continue to the next task.

5 Launching and Verifying an External Instance

Up to this point, you have learned how to create an external network, a router, a floating IP address, an SSH key pair, and a security group. These are all the resources necessary to create and interact with an external instance from outside the OpenStack cloud. In this task, you will put all these concepts together to create a functioning external instance. You will create the necessary resources, launch an external instance, and verify its connectivity and functionality with the **ssh** and **ping** commands.

- 5.1. If a terminal window is not already open, open one and source the admin credentials from the `~/keystonerc-admin` file.

```
ubuntu@workstation:~$ source ~/keystonerc-admin
```

```
ubuntu@workstation:~$ source ~/keystonerc-admin
[ubuntu@workstation (keystone-admin)]:~$
```

- 5.2. Before creating our own router and external network, we need to delete the ones that are created by default. First, unset the external gateway from **router1**.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack router unset \
> --external-gateway \
> router1
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack router unset \
> --external-gateway \
> router1
[ubuntu@workstation (keystone-admin)]:~$
```

- 5.3. List the interfaces of **router1**.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack port list \
> -c ID \
> -f value \
> --router router1
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack port list \
> -c ID \
> -f value \
> --router router1
a05f4e1c-4014-4d25-9538-64e8114197e1
d369b706-db4a-4239-b715-721708931870
[ubuntu@workstation (keystone-admin)]:~$
```

5.4. Capture the output of the previous command in to a variable called **ports**.

```
[ubuntu@workstation (keystone-admin)]:~$ ports=$(!!)
```

```
[ubuntu@workstation (keystone-admin)]:~$ ports=$(!!)
ports=$(openstack port list -c ID -f value --router router1)
[ubuntu@workstation (keystone-admin)]:~$ █
```

5.5. Ensure that **ports** contains the ID values as expected.

```
[ubuntu@workstation (keystone-admin)]:~$ echo $ports
```

```
[ubuntu@workstation (keystone-admin)]:~$ echo $ports
a05f4e1c-4014-4d25-9538-64e8114197e1 d369b706-db4a-4239-b715-721708931870
[ubuntu@workstation (keystone-admin)]:~$ █
```

5.6. Remove the interfaces from **router1**.

```
[ubuntu@workstation (keystone-admin)]:~$ for port in $ports; do \
> openstack router remove port router1 $port; \
> done
```

```
[ubuntu@workstation (keystone-admin)]:~$ for port in $ports; do \
> openstack router remove port router1 $port; \
> done
[ubuntu@workstation (keystone-admin)]:~$ █
```

5.7. Delete **router1**.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack router delete router1
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack router delete router1
[ubuntu@workstation (keystone-admin)]:~$ █
```

5.8. Delete the **public** network.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack network delete public
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack network delete public
[ubuntu@workstation (keystone-admin)]:~$ █
```

- 5.9. First, we will create an external network. List the existing networks to ensure that one does not already exist. Only the **shared** and **private** networks should be listed in the output.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack network list
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack network list
+-----+-----+-----+
| ID                               | Name   | Subnets |
+-----+-----+-----+
| 966ecb4f-4ff8-44ea-a476-2d2f18955085 | private | 674205b6-1357-4727-a21a-94220492a57f, fa8a2545-5a8c-44a2-bacc-1b86c253b880 |
| 9f23266f-d833-4337-9a27-4818a6d28e9e | shared  | 7e456257-76e5-4cfc-bf3f-b2a3876dba40 |
+-----+-----+-----+
[ubuntu@workstation (keystone-admin)]:~$
```

- 5.10.** Create an external network named **external**. Set the network type to **flat** and the physical network to **public**. Set the network as shared and external.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack network create \
> --external \
> --share \
> --provider-network-type flat \
> --provider-physical-network public \
> external
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack network create \
> --external \
> --share \
> --provider-network-type flat \
> --provider-physical-network public \
> external
```

Field	Value
admin_state_up	UP
availability_zone_hints	
availability_zones	
created_at	2025-07-02T15:57:18Z
description	
dns_domain	None
id	c231ef7c-399a-41ee-b95a-911d9a2abdbe
ipv4_address_scope	None
ipv6_address_scope	None
is_default	False
is_vlan_transparent	None
mtu	1500
name	external
port_security_enabled	True
project_id	39e851b14f864573aad60582c35e40dc
provider:network_type	flat
provider:physical_network	public
provider:segmentation_id	None
qos_policy_id	None
revision_number	1
router:external	External
segments	None
shared	True
status	ACTIVE
subnets	
tags	
updated_at	2025-07-02T15:57:18Z

```
[ubuntu@workstation (keystone-admin)]:~$
```

- 5.11.** Create a subnet named **external-subnet** in the **external** network. Give the subnet a range of **172.25.250.60** to **172.25.250.80**. Disable DHCP services for the subnet, and use the address **172.25.250.254** as the gateway and the DNS name server.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack subnet create \
> --subnet-range 172.25.250.0/24 \
> --no-dhcp \
> --gateway 172.25.250.254 \
> --dns-nameserver 172.25.250.254 \
> --allocation-pool start=172.25.250.60,end=172.25.250.80 \
> --network external \
> external-subnet
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack subnet create \
> --subnet-range 172.25.250.0/24 \
> --no-dhcp \
> --gateway 172.25.250.254 \
> --dns-nameserver 172.25.250.254 \
> --allocation-pool start=172.25.250.60,end=172.25.250.80 \
> --network external \
> external-subnet
+-----+-----+
| Field | Value |
+-----+-----+
| allocation_pools | 172.25.250.60-172.25.250.80 |
| cidr | 172.25.250.0/24 |
| created_at | 2025-07-02T15:58:58Z |
| description | |
| dns_nameservers | 172.25.250.254 |
| enable_dhcp | False |
| gateway_ip | 172.25.250.254 |
| host_routes | |
| id | 1e2fe0a9-5ed4-4e00-aeb5-6040164f6bdf |
| ip_version | 4 |
| ipv6_address_mode | None |
| ipv6_ra_mode | None |
| name | external-subnet |
| network_id | c231ef7c-399a-41ee-b95a-911d9a2abdbe |
| project_id | 39e851b14f864573aad60582c35e40dc |
| revision_number | 0 |
| segment_id | None |
| service_types | |
| subnetpool_id | None |
| tags | |
| updated_at | 2025-07-02T15:58:58Z |
+-----+-----+
[ubuntu@workstation (keystone-admin)]:~$
```

5.12. List the networks again to verify the creation of the **external** network.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack network list
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack network list
+-----+-----+-----+
| ID                                     | Name   | Subnets |
+-----+-----+-----+
| 966ecb4f-4ff8-44ea-a476-2d2f18955085 | private | 674205b6-1357-4727-a21a-94220492a57f, fa8a2545-5a8c-44a2-bacc-1b86c253b880 |
| 9f23266f-d833-4337-9a27-4818a6d28e9e | shared  | 7e456257-76e5-4cfc-bf3f-b2a3876dba40 |
| c231ef7c-399a-41ee-b95a-911d9a2abdbe | external | 1e2fe0a9-5ed4-4e00-aeb5-6040164f6bdf |
+-----+-----+-----+
[ubuntu@workstation (keystone-admin)]:~$
```

5.13. List the existing routers. The output should be empty.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack router list
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack router list
[ubuntu@workstation (keystone-admin)]:~$
```

5.14. Next, create a router named **router-external**.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack router create router-external
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack router create router-external
+-----+-----+
| Field                | Value |
+-----+-----+
| admin_state_up       | UP    |
| availability_zone_hints |      |
| availability_zones    |      |
| created_at           | 2025-07-02T16:00:14Z |
| description          |      |
| distributed           | False |
| external_gateway_info | None  |
| flavor_id            | None  |
| ha                    | False |
| id                   | f6d6293d-86cf-4c7b-bab4-07cca9757d2e |
| name                 | router-external |
| project_id           | 39e851b14f864573aad60582c35e40dc |
| revision_number       | 1     |
| routes               |      |
| status               | ACTIVE |
| tags                 |      |
| updated_at           | 2025-07-02T16:00:14Z |
+-----+-----+
[ubuntu@workstation (keystone-admin)]:~$
```


5.15. List the routers again to verify the creation of **router-external**.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack router list
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack router list
+-----+-----+-----+-----+-----+-----+-----+
| ID          | Name      | Status | State | Distributed | HA    | Project          |
+-----+-----+-----+-----+-----+-----+-----+
| f6d6293d-86cf-4c7b-bab4-07cca9757d2e | router-external | ACTIVE | UP    | False       | False | 39e851b14f864573aad60582c35e40dc |
+-----+-----+-----+-----+-----+-----+-----+
[ubuntu@workstation (keystone-admin)]:~$
```

5.16. The router needs to be connected to both the **shared-subnet** and **external-subnet** networks to allow external connections to the instance. First, add the **shared-subnet** to the router.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack router add subnet \
> router-external \
> shared-subnet
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack router add subnet \
> router-external \
> shared-subnet
[ubuntu@workstation (keystone-admin)]:~$
```

5.17. Set the **external** network as the gateway for the router.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack router set \
> --external-gateway external \
> router-external
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack router set \
> --external-gateway external \
> router-external
[ubuntu@workstation (keystone-admin)]:~$
```

5.18. Show the details of **router-external** to show that it is connected to two subnets.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack router show router-external
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack router show router-external
+-----+-----+
| Field | Value |
+-----+-----+
| admin_state_up | UP |
| availability_zone_hints |  |
| availability_zones |  |
| created_at | 2025-07-02T16:00:14Z |
| description |  |
| distributed | False |
| external_gateway_info | {"network_id": "c231ef7c-399a-41ee-b95a-911d9a2abdbe", "enable_snat": true, "external_fixed_ips": [{"subnet_id": "1e2fe0a9-5ed4-4e00-aeb5-6040164f6bdf", "ip_address": "172.25.250.76"}]} |
| flavor_id | None |
| ha | False |
| id | f6d6293d-86cf-4c7b-bab4-07cca9757d2e |
| interfaces_info | [{"subnet_id": "7e456257-76e5-4cfc-bf3f-b2a3876dba40", "ip_address": "192.168.233.1", "port_id": "9288a863-28c3-421a-9a5e-7346d30ea66c"}] |
| name | router-external |
| project_id | 39e851b14f864573aad60582c35e40dc |
| revision_number | 4 |
| routes |  |
| status | ACTIVE |
| tags |  |
| updated_at | 2025-07-02T16:01:46Z |
+-----+-----+
[ubuntu@workstation (keystone-admin)]:~$
```

5.19. Now, we will create a floating IP address to associate with the instance. This will be the IP address we use to connect to the instance through the **external** network. First, list the available floating IP addresses. The output should be empty.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack floating ip list
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack floating ip list
```

5.20. Create the floating IP address **172.25.250.75** in the **external** network.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack floating ip create \
> --floating-ip-address 172.25.250.75 \
> external
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack floating ip create \
> --floating-ip-address 172.25.250.75 \
> external
```

Field	Value
created_at	2025-07-02T16:03:11Z
description	
fixed_ip_address	None
floating_ip_address	172.25.250.75
floating_network_id	c231ef7c-399a-41ee-b95a-911d9a2abdbe
id	845f62f4-c0bc-4ba4-989c-f42bf60ed3cc
name	172.25.250.75
port_id	None
project_id	39e851b14f864573aad60582c35e40dc
qos_policy_id	None
revision_number	0
router_id	None
status	DOWN
subnet_id	None
updated_at	2025-07-02T16:03:11Z

```
[ubuntu@workstation (keystone-admin)]:~$
```

5.21. List the floating IP addresses again to verify the creation of **172.25.250.75**.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack floating ip list
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack floating ip list
```

ID	Floating IP Address	Fixed IP Address	Port	Floating Network	Project
845f62f4-c0bc-4ba4-989c-f42bf60ed3cc	172.25.250.75	None	None	c231ef7c-399a-41ee-b95a-911d9a2abdbe	39e851b14f864573aad60582c35e40dc

```
[ubuntu@workstation (keystone-admin)]:~$
```

- 5.22. In order for our ICMP and SSH requests to reach our external instance, we need to set up a security group and security rules that allow this traffic. First, list the available security groups. The output should contain two default security groups.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack security group list
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack security group list
```

ID	Name	Description	Project
2f0f5133-8396-45ea-a4de-61945d79ed2e	default	Default security group	eb2dcd08d8ae46ffac3f16c3973ef61d
62dafb67-e7fd-44d6-bf87-f4701dd66875	default	Default security group	39e851b14f864573aad60582c35e40dc

```
[ubuntu@workstation (keystone-admin)]:~$
```

- 5.23. Create a security group. Because we will allow ICMP and SSH traffic, we will name the security group **sg-allow-icmp-ssh**.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack security group create \
> sg-allow-icmp-ssh
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack security group create \
> sg-allow-icmp-ssh
```

Field	Value
created_at	2025-07-02T16:09:59Z
description	sg-allow-icmp-ssh
id	ed0d9549-6476-4a35-a2d1-24f140db7c73
name	sg-allow-icmp-ssh
project_id	39e851b14f864573aad60582c35e40dc
revision_number	1
rules	created_at='2025-07-02T16:09:59Z', direction='egress', ethertype='IPv4', id='0b2244a9-b588-40e4-aaa-1ee04281d927', standard_attr_id='54', updated_at='2025-07-02T16:09:59Z'
	created_at='2025-07-02T16:09:59Z', direction='egress', ethertype='IPv6', id='f925dbf8-9bc9-48b5-92f2-64106c8f34f5', standard_attr_id='53', updated_at='2025-07-02T16:09:59Z'
updated_at	2025-07-02T16:09:59Z

```
[ubuntu@workstation (keystone-admin)]:~$
```

Tip

Real-world OpenStack environments may have a variety of instances with different security group needs. Therefore, it pays to be descriptive in your naming conventions so that you can more easily identify the appropriate security group later. It is much easier to tell what **sg-allow-icmp-ssh** does than it is to guess the purpose of **secgroup1**. However, giving strong names alone is not enough—it is always best to check a security group's actual rules before adding it to an instance.

5.24. List the security groups to verify the creation of **sg-allow-icmp-ssh**.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack security group list
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack security group list
```

ID	Name	Description	Project
2f0f5133-8396-45ea-a4de-61945d79ed2e	default	Default security group	eb2dcd08d8ae46ffa-c3f16c3973ef61d
62dafb67-e7fd-44d6-bf87-f4701dd66875	default	Default security group	39e851b14f864573a-ad60582c35e40dc
ed0d9549-6476-4a35-a2d1-24f140db7c73	sg-allow-icmp-ssh	sg-allow-icmp-ssh	39e851b14f864573a-ad60582c35e40dc

```
[ubuntu@workstation (keystone-admin)]:~$
```

5.25. List the rules in the **sg-allow-icmp-ssh** security group. There should be two rules created by default.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack security group rule list \
> sg-allow-icmp-ssh
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack security group rule list \
> sg-allow-icmp-ssh
```

ID	IP Protocol	IP Range	Port Range	Remote Security Group
0b2244a9-b588-40e4-aeea-1ee04281d927	None	None		None
f925dbf8-9bc9-48b5-92f2-64106c8f34f5	None	None		None

```
[ubuntu@workstation (keystone-admin)]:~$
```

Tip

Try to use the **for** loop trick to verify that the default security rules allow IPv4 and IPv6 egress traffic. You can use this command to show the details of a rule:

```
openstack security group rule show <ID>
```

- 5.26. Add a security rule in the **sg-allow-icmp-ssh** security group to allow all incoming ICMP traffic.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack security group rule create \
> --protocol icmp \
> sg-allow-icmp-ssh
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack security group rule create \
> --protocol icmp \
> sg-allow-icmp-ssh
```

Field	Value
created_at	2025-07-02T16:11:55Z
description	
direction	ingress
ether_type	IPv4
id	deefbca4-8ba4-4ee8-9131-6b8e8d2903b7
name	None
port_range_max	None
port_range_min	None
project_id	39e851b14f864573aad60582c35e40dc
protocol	icmp
remote_group_id	None
remote_ip_prefix	0.0.0.0/0
revision_number	0
security_group_id	ed0d9549-6476-4a35-a2d1-24f140db7c73
updated_at	2025-07-02T16:11:55Z

```
[ubuntu@workstation (keystone-admin)]:~$
```

- 5.27. Add another security rule to allow remote connection using SSH on the default port 22.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack security group rule create \
> --protocol tcp \
> --dst-port 22 \
> sg-allow-icmp-ssh
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack security group rule create \
> --protocol tcp \
> --dst-port 22 \
> sg-allow-icmp-ssh
```

Field	Value
created_at	2025-07-02T16:12:48Z
description	
direction	ingress
ether_type	IPv4
id	6df08e02-9958-4e04-9603-e4ce8caa3f49
name	None
port_range_max	22
port_range_min	22
project_id	39e851b14f864573aad60582c35e40dc
protocol	tcp
remote_group_id	None
remote_ip_prefix	0.0.0.0/0
revision_number	0
security_group_id	ed0d9549-6476-4a35-a2d1-24f140db7c73
updated_at	2025-07-02T16:12:48Z

```
[ubuntu@workstation (keystone-admin)]:~$
```

- 5.28. List the rules of the **sg-allow-icmp-ssh** security group again to verify that there are now four rules: the two default rules plus the two rules you just added.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack security group rule list \
> sg-allow-icmp-ssh
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack security group rule list \
> sg-allow-icmp-ssh
```

ID	IP Protocol	IP Range	Port Range	Remote Security Group
0b2244a9-b588-40e4-aeaa-1ee04281d927	None	None		None
6df08e02-9958-4e04-9603-e4ce8caa3f49	tcp	0.0.0.0/0	22:22	None
deefbca4-8ba4-4ee8-9131-6b8e8d2903b7	icmp	0.0.0.0/0		None
f925dbf8-9bc9-48b5-92f2-64106c8f34f5	None	None		None

```
[ubuntu@workstation (keystone-admin)]:~$
```

- 5.29. Finally, we need an SSH key pair to remotely log in to the external instance. List the available key pairs. The output should be empty.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack keypair list
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack keypair list
[ubuntu@workstation (keystone-admin)]:~$
```

- 5.30. Create a key pair called **keypair**, and save the private key to **/Downloads/keypair.pem**.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack keypair create \
> keypair > ~/Downloads/keypair.pem
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack keypair create \
> keypair > ~/Downloads/keypair.pem
[ubuntu@workstation (keystone-admin)]:~$
```

- 5.31. Set the file permissions of the private key so that only the **ubuntu** user has read/write permissions.

```
[ubuntu@workstation (keystone-admin)]:~$ chmod 600 ~/Downloads/keypair.pem
```

```
[ubuntu@workstation (keystone-admin)]:~$ chmod 600 ~/Downloads/keypair.pem
[ubuntu@workstation (keystone-admin)]:~$
```


5.32. List the available key pairs again to verify the creation of **keypair**.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack keypair list
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack keypair list
+-----+-----+
| Name   | Fingerprint |
+-----+-----+
| keypair | 66:30:11:e1:98:80:8c:47:d2:a9:f0:c9:62:5e:4a:33 |
+-----+-----+
[ubuntu@workstation (keystone-admin)]:~$
```

5.33. We now have all the resources we need to create an external instance. List all instances in the project. The output should be empty.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack server list
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack server list
[ubuntu@workstation (keystone-admin)]:~$
```


- 5.34. Launch an instance named **instance-external** with the **ubuntu** image, the **m1.small** flavor, the **keypair2** key pair, the **shared** network, and the **secgroup2** security group.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack server create \
> --image ubuntu \
> --flavor m1.small \
> --network shared \
> --security-group sg-allow-icmp-ssh \
> --key-name keypair \
> instance-external
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack server create \
> --image ubuntu \
> --flavor m1.small \
> --network shared \
> --security-group sg-allow-icmp-ssh \
> --key-name keypair \
> instance-external
```

Field	Value
OS-DCF:diskConfig	MANUAL
OS-EXT-AZ:availability_zone	
OS-EXT-SRV-ATTR:host	None
OS-EXT-SRV-ATTR:hypervisor_hostname	None
OS-EXT-SRV-ATTR:instance_name	
OS-EXT-STS:power_state	NOSTATE
OS-EXT-STS:task_state	scheduling
OS-EXT-STS:vm_state	building
OS-SRV-USG:launched_at	None
OS-SRV-USG:terminated_at	None
accessIPv4	
accessIPv6	
addresses	
adminPass	DM33HA3CWcci
config_drive	
created	2025-07-02T16:16:30Z
flavor	m1.small (2)
hostId	
id	174dbcb2-0d5f-4f19-b6f2-729d8dbaa540
image	ubuntu (329d361e-f6dc-4b72-b200-3de0ec230e65)
key_name	keypair
name	instance-external
progress	0
project_id	39e851b14f864573aad60582c35e40dc
properties	
security_groups	name='ed0d9549-6476-4a35-a2d1-24f140db7c73'
status	BUILD
updated	2025-07-02T16:16:29Z
user_id	14f5376f00c04e90b7103dd8d4263040
volumes_attached	

```
[ubuntu@workstation (keystone-admin)]:~$
```

5.35. List the floating IPs.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack floating ip list
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack floating ip list
+-----+-----+-----+-----+-----+-----+
| ID | Floating IP Address | Fixed IP Address | Port | Floating Network | Project |
+-----+-----+-----+-----+-----+-----+
| 845f62f4-c0bc-4ba4-989c-f42bf60ed3cc | 172.25.250.75 | None | None | c231ef7c-399a-41ee-b95a-911d9a2abdbe | 39e851b14f864573aad60582c35e40dc |
+-----+-----+-----+-----+-----+-----+
[ubuntu@workstation (keystone-admin)]:~$
```

5.36. Associate the floating IP address with the **instance-external** instance.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack server add floating ip \
> instance-external \
> 172.25.250.75
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack server add floating ip \
> instance-external \
> 172.25.250.75
[ubuntu@workstation (keystone-admin)]:~$
```

5.37. Verify that the instance was assigned the floating IP address.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack server list \
> -c Name \
> -c Networks
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack server list \
> -c Name \
> -c Networks
+-----+-----+-----+
| Name | Networks |
+-----+-----+-----+
| instance-external | shared=192.168.233.230, 172.25.250.75 |
+-----+-----+-----+
[ubuntu@workstation (keystone-admin)]:~$
```

5.38. Use the **scp** command to send the **keypair2** key pair to the **devstack** machine over SSH. Use the password **ubuntu** for authentication.

```
[ubuntu@workstation (keystone-admin)]:~$ scp ~/Downloads/keypair.pem \
> 192.168.1.20:~/keypair.pem
```

```
[ubuntu@workstation (keystone-admin)]:~$ scp ~/Downloads/keypair.pem \
> 192.168.1.20:~/keypair.pem
ubuntu@192.168.1.20's password:
keypair.pem 100% 1680 2.9MB/s 00:00
[ubuntu@workstation (keystone-admin)]:~$
```

Note

Because your username is **ubuntu** on both machines, you don't need to specify the username in the **scp** command. By default, **scp** uses your current local username for the remote connection. If you wanted to send the file to a different user, such as **user2** on the **devstack** machine, you would write: **user2@192.168.1.20:~/keypair.pem**.

5.39. SSH into the **devstack** machine. Use the password **ubuntu** when prompted.

```
[ubuntu@workstation (keystone-admin)]:~$ ssh 192.168.1.20
```

```
[ubuntu@workstation (keystone-admin)]:~$ ssh 192.168.1.20
ubuntu@192.168.1.20's password:
Welcome to Ubuntu 22.04.3 LTS (GNU/Linux 5.15.0-94-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

This system has been minimized by removing packages and content that are
not required on a system that users do not log into.

To restore this content, you can run the 'unminimize' command.
Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your I
nternet connection or proxy settings

Last login: Thu Oct 17 01:27:44 2024
ubuntu@devstack:~$
```

5.40. Ping the **instance-external** instance with the floating IP that was assigned to it in the previous task. This will verify that the instance can be reached and that its security group allows ICMP ingress traffic.

```
ubuntu@devstack:~$ ping -c3 172.25.250.75
```

```
ubuntu@devstack:~$ ping -c3 172.25.250.75
PING 172.25.250.75 (172.25.250.75) 56(84) bytes of data.
64 bytes from 172.25.250.75: icmp_seq=1 ttl=63 time=22.4 ms
64 bytes from 172.25.250.75: icmp_seq=2 ttl=63 time=2.28 ms
64 bytes from 172.25.250.75: icmp_seq=3 ttl=63 time=1.44 ms

--- 172.25.250.75 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2003ms
rtt min/avg/max/mdev = 1.441/8.700/22.383/9.681 ms
ubuntu@devstack:~$
```

Note

You should receive three successful ping replies.

- 5.41.** SSH into the **instance-external** instance with the **keypair2.pem** file. Enter **yes** when asked if you want to continue.

```
ubuntu@devstack:~$ ssh -i ~/keypair2.pem \
> 172.25.250.75
```

```
ubuntu@devstack:~$ ssh -i ~/keypair.pem \
> 172.25.250.75
The authenticity of host '172.25.250.75 (172.25.250.75)' can't be established.
ED25519 key fingerprint is SHA256:u75EWhnyJ8tc8P2xb3bIrBIT0Dw+9uEhXahNpjuG8/o.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '172.25.250.75' (ED25519) to the list of known hosts.
Welcome to Ubuntu 22.04.3 LTS (GNU/Linux 5.15.0-92-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Wed Jul  2 16:27:20 UTC 2025

System load:  0.2216796875      Processes:           85
Usage of /:   7.4% of 19.20GB   Users logged in:    0
Memory usage: 8%               IPv4 address for ens3: 192.168.233.230
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@instance-external:~$
```

Note

It may take a few minutes for the instance to be fully booted and ready to accept SSH connections.

Note

It is important to connect to the instance through SSH from the **devstack** machine since it is outside the OpenStack cloud. A successful connection verifies the external connectivity of the instance.

5.42. Ping the DHCP server on the **shared** network to verify connectivity.

```
ubuntu@instance-external:~$ ping -c3 192.168.233.2
```

```
ubuntu@instance-external:~$ ping -c3 192.168.233.2
PING 192.168.233.2 (192.168.233.2) 56(84) bytes of data.
64 bytes from 192.168.233.2: icmp_seq=1 ttl=64 time=5.15 ms
64 bytes from 192.168.233.2: icmp_seq=2 ttl=64 time=2.67 ms
64 bytes from 192.168.233.2: icmp_seq=3 ttl=64 time=0.729 ms

--- 192.168.233.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2005ms
rtt min/avg/max/mdev = 0.729/2.852/5.154/1.810 ms
ubuntu@instance-external:~$
```

Note

You should receive three successful ping replies.

5.43. The lab is now complete.

A Fine-Grained Security Group Control

In Section 2, we discussed applying security groups to instances. However, in reality, security groups are applied to the individual ports on an instance. For simplicity, we could mostly ignore this distinction and treat them as if they applied directly to the instance. More fine-grained control over security groups is available from both the CLI and web interface, which is especially useful when working with instances that have multiple interfaces that each have different requirements.

From the dashboard, managing security groups for individual ports only requires navigating a new menu. In Section 2, Step 18, we clicked **Edit Security Groups** in the instance's dropdown menu. To manage security groups for a specific interface instead, click **Edit Port Security Groups**. The interface is nearly identical.

From the CLI, working with individual ports requires slightly different commands. The following commands apply to the *entire* instance—that is, to all its ports (with some exceptions):

```
openstack server add security group <instance> <security_group>
openstack server remove security group <instance> <security_group>
```

Security groups can also be managed for each port from the CLI; however, the process can be slightly more cumbersome because unnamed ports must be specified by ID.

First, it is recommended to maximize your terminal window to prevent ID values from being split across lines. This will make it easier to copy the IDs to the clipboard. View the ports of an instance to get their IDs.

```
openstack port list --server <instance_name_or_id>
```

For each port you want to modify, first copy its ID to the clipboard. Alternatively, use the ID to name the port with

```
openstack port set --name <new_port_name> <port_name_or_id>
```

or name a port upon creation with

```
openstack port create --name <port_name> --network <network_name_or_id>
```

Next, find the security groups applied to the port.

```
openstack port show <port_name_or_id> -c security_group_ids
```

Optionally, find the name of each security group.

```
openstack security group show <security_group_id> -c name
```

Finally, remove a security group from an interface individually with

```
openstack port unset --security-group <security_group_name_or_id> <port_name_or_id>
```

or remove all security groups from an interface with

```
openstack port set --no-security-group <port_name_or_id>
```